

VIETNAM GENERAL CONFEDERATION OF LABOR

TON DUC THANG UNIVERSITY

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RESEARCH TOPIC 1

DETECTION OF DEPRESSION SIGNS IN VIETNAMESE SOCIAL MEDIA TEXT USING DEEP LEARNING MODELS

HO CHI MINH CITY, 2026

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Ho Chi Minh city, 10th March 2026

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PHÁT HIỆN DẤU HIỆU TRẦM CẢM TRONG VĂN BẢN TIẾNG VIỆT TRÊN MẠNG XÃ HỘI SỬ DỤNG CÁC MÔ HÌNH HỌC SÂU

TÓM TẮT

Trầm cảm là một trong những rối loạn tâm thần phổ biến nhất trên thế giới, ảnh hưởng đến khoảng 280 triệu người theo thống kê của Tổ chức Y tế Thế giới (WHO). Việc phát hiện sớm các dấu hiệu trầm cảm đóng vai trò quan trọng trong việc ngăn ngừa và điều trị kịp thời căn bệnh này. Trong bối cảnh mạng xã hội ngày càng phát triển, YouTube đã trở thành một nền tảng chia sẻ nội dung phổ biến, nơi người dùng Việt Nam thường xuyên bày tỏ cảm xúc và tâm trạng của mình thông qua các bình luận. Tuy nhiên, việc nghiên cứu phát hiện trầm cảm dựa trên dữ liệu bình luận YouTube tiếng Việt vẫn còn rất hạn chế.

Trong luận văn này, chúng tôi giới thiệu một nghiên cứu toàn diện đầu tiên về phát hiện dấu hiệu trầm cảm từ bình luận YouTube tiếng Việt sử dụng các phương pháp học sâu (Deep Learning). Đóng góp chính của nghiên cứu bao gồm việc xây dựng bộ dữ liệu VietDepression với 125.329 bình luận YouTube được thu thập từ các kênh Việt Nam, trong đó bộ dữ liệu vàng (golden dataset) gồm 1.607 mẫu đã được gán nhãn thủ công và xác minh bởi người để đảm bảo độ chính xác cao.

Để giải quyết vấn đề thiếu nhãn trong dữ liệu quy mô lớn, chúng tôi đề xuất một hệ thống weak labeling sử dụng 335 từ khóa liên quan đến trầm cảm, đạt được khả năng phủ sóng 98% recall và độ chính xác 61% precision. Hệ thống này cho phép mở rộng quy mô dữ liệu huấn luyện một cách hiệu quả. Chúng tôi tiến hành đánh giá và so sánh năm mô hình khác nhau, bao gồm: TF-IDF kết hợp LinearSVC, BiLSTM, PhoBERT, BERTopic, và mô hình kết hợp PhoBERT với BERTopic.

Kết quả thực nghiệm cho thấy mô hình PhoBERT đạt hiệu suất tốt nhất với chỉ số F1-macro đạt 0.7187 trên tập dữ liệu in-domain và 0.3798 trên tập dữ liệu cross-domain. Các kết quả này chứng minh tính khả thi của việc áp dụng các mô hình ngôn ngữ tiên huấn luyện cho bài toán phát hiện trầm cảm trong ngữ cảnh tiếng Việt, đồng thời mở ra hướng nghiên cứu tiềm năng trong lĩnh vực chăm sóc sức khỏe tinh thần dựa trên mạng xã hội.

DETECTION OF DEPRESSION SIGNS IN VIETNAMESE SOCIAL MEDIA TEXT USING DEEP LEARNING MODELS

ABSTRACT

Depression is one of the most prevalent mental disorders worldwide, affecting approximately 280 million people according to the World Health Organization (WHO). Early detection of depression signs plays a crucial role in preventing and treating this condition. With the rapid growth of social media, YouTube has become a popular content-sharing platform where Vietnamese users frequently express their emotions and moods through comments. However, research on depression detection based on Vietnamese YouTube comment data remains very limited.

In this thesis, we present a comprehensive first-of-its-kind study on detecting depression signs from Vietnamese YouTube comments using Deep Learning methods. The main contributions include the construction of the VietDepression dataset with 125,329 YouTube comments collected from Vietnamese channels, where a golden dataset of 1,607 samples was manually labeled and verified to ensure high accuracy.

To address the label scarcity problem in large-scale data, we propose a weak labeling system using 335 depression-related keywords, achieving 98% recall coverage and 61% precision accuracy. This system enables effective scaling of training data. We conducted evaluations and comparisons of five different models: TF-IDF combined with LinearSVC, BiLSTM, PhoBERT, BERTopic, and a combined PhoBERT with BERTopic model.

Experimental results show that the PhoBERT model achieves the best performance with an F1-macro score of 0.7187 on the in-domain dataset and 0.3798 on the cross-domain dataset. These results demonstrate the feasibility of applying pre-trained language models for depression detection in Vietnamese context, while also opening promising research directions in mental health care based on social media.

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ABBREVIATIONS

API	Application Programming Interface
BiLSTM	Bidirectional Long Short-Term Memory
BERT	Bidirectional Encoder Representations from Transformers
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, 5th Edition
DAPT	Domain-Adaptive Pre-training
LIWC	Linguistic Inquiry and Word Count
NLP	Natural Language Processing
PHQ-9	Patient Health Questionnaire-9
PhoBERT	Pre-trained BERT for Vietnamese
TF-IDF	Term Frequency-Inverse Document Frequency
VSMEC	Vietnamese Social Media Emotion Classification

CHAPTER 1: INTRODUCTION

1.1 Motivation and Problem Statement

Depression (Major Depressive Disorder) is one of the most prevalent mental health disorders globally, affecting approximately 280 million people according to the World Health Organization (WHO, 2023). Beyond the personal suffering it causes, depression carries substantial economic burden through reduced productivity, healthcare costs, and in severe cases, suicide-related losses. In Vietnam, the prevalence of depressive disorders has been steadily increasing, particularly among adolescents and young adults aged 15-24, who represent a demographic particularly vulnerable to mental health challenges while also constituting the largest group of social media users in the country [24].

While clinical screening tools such as PHQ-9 (Patient Health Questionnaire-9) and DSM-5 (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition) provide standardized methods for depression assessment [1], the majority of individuals who need professional help do not receive it. This treatment gap is attributed to multiple factors: social stigma surrounding mental health issues that discourages individuals from seeking help, limited public awareness about depression symptoms and their clinical significance, and a severe shortage of mental health professionals. In Vietnam, there are approximately only 200 psychiatrists serving a population of 100 million, yielding a psychiatrist-to-population ratio far below World Health Organization recommendations [24].

Social media platforms, especially YouTube, have emerged as a rich source of naturally occurring Vietnamese text where users openly discuss their emotional states, personal struggles, and mental health concerns without the barriers of stigma or access. These public expressions provide an opportunity to study early signs of depression through language analysis without requiring individuals to actively seek clinical help [4]. This approach, known as computational psychiatry or social media-based mental health detection, has gained significant research attention over the past decade [9, 10].

The Five Depression Signs We Target: In this research, we focus on detecting five core signs of depression as defined by clinical diagnostic criteria:

1. **Prolonged Sadness (Buồn bã kéo dài):** Persistent feelings of sadness, emptiness, or feeling down that persist over time, as manifested in Vietnamese expressions such as “*buồn quá,*” “*buồn không có lý do,*” or “*nặng trĩu con tim.*”
2. **Loss of Motivation (Mất động lực):** Anhedonia or loss of interest in activities once enjoyed, reduced drive to engage with work, study, or social activities, as seen in comments like “*không muốn làm gì cả,*” “*chán mọi thứ,*” or “*không có động lực để sống tiếp.*”
3. **Hopelessness (Tuyệt vọng):** Feelings of despair, pessimism about the future, or belief that situations will not improve, expressed through phrases such as “*cuộc đời này vô nghĩa,*” “*không còn gì để mong chờ,*” or “*mọi thứ đều vô ích.*”
4. **Social Isolation (Tự cô lập):** Withdrawal from friends, family, and social activities, preferring solitude or expressing feelings of loneliness, as in “*muốn ở một mình,*” “*không muốn gặp ai,*” or “*cô đơn lắm rồi.*”
5. **Negative Thoughts (Ý nghĩ tiêu cực):** Recurrent negative thinking patterns, self-criticism, feelings of worthlessness, or excessive guilt, manifesting in comments like “*mình vô dụng,*” “*tội lỗi gì đâu,*” or “*ai cũng giỏi hơn mình.*”

These five signs align with the diagnostic criteria for Major Depressive Disorder as defined by DSM-5 [1] and represent the most commonly expressed depression indicators in Vietnamese social media text.

The Opportunity of Vietnamese Social Media: Vietnam has one of the highest social media usage rates globally, with approximately 70 million active social media users as of 2023. YouTube, in particular, functions almost like a social network in Vietnam, where users engage in long, personal comment discussions on videos that resonate with their emotional states. This creates a unique opportunity to analyze authentic expressions of mental health concerns in Vietnamese.

However, developing automated systems for detecting these five depression signs in Vietnamese social media text presents several unique challenges [3, 25]:

- **Domain-specific dataset scarcity:** While depression detection has been extensively studied in English with numerous labeled datasets (e.g., SMHD, eRisk), there is a critical lack of Vietnamese depression detection datasets with human-annotated labels suitable for training deep learning models. Existing Vietnamese NLP datasets focus on general sentiment or emotion recognition, not the specific depression indicators we target.
- **Linguistic challenges:** Vietnamese is a tonal language with six distinct tones marked by diacritic signs. The language exhibits complex word-compounding structures where most words consist of single syllables combined into compounds. Social media text presents additional challenges: informal language with missing diacritics (“*buon*” instead of “*buồn*”), teen slang (“*vgl*,” “*vl*,” and abbreviations “*ko*,” “*dc*,” “*j*”), mixed code-switching with English, and expressive lengthening (“*buồnnnnnnn*”). These informal patterns differ significantly from the formal Vietnamese text used to pre-train existing language models.
- **Label quality:** Creating high-quality labeled datasets for depression detection requires careful annotation protocols that distinguish between clinical depression indicators and normal emotional expressions. This makes data collection both time-consuming and expensive, necessitating semi-supervised approaches such as weak labeling and active learning.
- **Generalization concerns:** Models trained on YouTube comments from one demographic may struggle to generalize to other platforms (Facebook, TikTok) or domains (student essays, formal writing), creating a significant generalization gap that limits real-world applicability. This is particularly relevant given the different linguistic registers between social media and clinical text.

1.2 Research Gap

Despite significant advances in depression detection research in English, several critical gaps remain in the literature, particularly for Vietnamese NLP:

1. **Lack of Vietnamese depression-specific datasets:** No publicly available Vietnamese dataset exists specifically for depression detection in social media text. Existing Vietnamese datasets such as UIT-VSMEC (emotion recognition), NTC-SCV (sentiment analysis), and UIT-VSFC (student feedback) [17] focus on general sentiment or emotion, not the five specific depression indicators we target: prolonged sadness, loss of motivation, hopelessness, social isolation, and negative thoughts.
2. **No comprehensive annotation pipelines:** While weak labeling and active learning have been successfully applied in other domains such as named entity recognition and sentiment analysis, no study has systematically explored their application for Vietnamese depression detection with blind human annotation protocols that ensure independent annotator judgments.
3. **Limited cross-domain evaluation:** Most existing depression detection studies evaluate models only on in-domain test sets drawn from the same distribution as training data. The generalization gap when models are applied to different platforms or domains remains underexplored in Vietnamese NLP, limiting our understanding of model transferability.
4. **No benchmarking of deep learning models:** Although PhoBERT and other deep learning models have shown state-of-the-art performance in Vietnamese NLP tasks, no comprehensive benchmark exists comparing multiple architectures specifically for depression detection, leaving practitioners without guidance on optimal model selection.

These gaps motivate our research to establish the first Vietnamese depression detection benchmark with comprehensive evaluation protocols.

1.3 Our Approach

To address the identified research gaps, we propose a comprehensive framework for detecting depression signs in Vietnamese YouTube comments using deep learning models. Our approach encompasses four key components:

- **Large-Scale Dataset Construction:** We construct the first Vietnamese depression detection dataset by collecting **125,329 Vietnamese YouTube comments** (exceeding the 100,000 minimum requirement) using 264 depression-related keywords organized in five thematic groups aligned with our target depression signs. We apply a three-level weighted keyword scoring system for weak labeling and conduct six rounds of active learning with blind annotation protocols to build a high-quality gold standard.
- **Semi-Supervised Learning Pipeline:** We develop a comprehensive weak-labeling and active learning pipeline that maximizes the value of limited human annotation resources. Our three-level weighted keyword system (335 entries) achieves 98% recall in identifying depression-related content, enabling us to scale from a small gold set to a larger training corpus.
- **Multi-Architecture Benchmarking:** We evaluate five representative deep learning models spanning different paradigms: TF-IDF + LinearSVC (traditional baseline), BiLSTM (pre-Transformer deep learning), PhoBERT (state-of-the-art Vietnamese language model), BERTopic (topic-based features), and our proposed PhoBERT + BERTopic hybrid approach.
- **Cross-Domain Evaluation:** We conduct comprehensive evaluation on both in-domain test sets and cross-domain test sets using the UIT-VSMEC dataset, enabling us to quantify the generalization gap and identify factors affecting model transferability.

fiX ME! – Proposed framework overview

Figure 1.1: Our proposed framework for Vietnamese depression detection: from data collection through model training to cross-domain evaluation. The pipeline comprises four main stages: dataset construction, weak labeling, active learning, and multi-model benchmarking.

1.4 Contributions

In this research, we make the following main contributions:

1. **We construct the first Vietnamese depression detection dataset** with both weak labels and human-annotated labels. The VietDepression dataset contains 125,329 YouTube comments with a 1,607-sample gold subset, providing a valuable benchmark for future Vietnamese mental health NLP research.
2. **We propose a comprehensive weak-labeling and active learning pipeline** for Vietnamese depression detection, incorporating a three-level weighted keyword system and six rounds of blind human annotation. This pipeline achieves 98% recall and 61% precision in detecting depression-related content.
3. **We conduct extensive experiments with five deep learning models** on both in-domain and cross-domain test sets, providing the first published benchmark results for Vietnamese depression detection. Our best model, PhoBERT, achieves F1-macro of 0.7187 on in-domain and 0.3608 on cross-domain evaluation.
4. **We provide a detailed analysis of the generalization gap** ($\Delta F1 \approx 0.3579$) between in-domain and cross-domain settings, identifying key factors that affect model transferability and suggesting directions for future improvement.
5. **We release a reusable Vietnamese weak-labeling word list** (335 entries across three levels) that can be adapted for other Vietnamese mental health text classification tasks.

1.5 Scope and Limitations

1.5.1 Scope

- **Platform:** We collect data only from YouTube. Other platforms (Facebook, TikTok, Zalo) are not included due to API limitations.
- **Language:** The system processes only Vietnamese text. Mixed Vietnamese-English text is handled without special preprocessing.
- **Classification:** We treat this as a binary classification problem (depression vs. normal). We do not assess severity levels or provide clinical diagnoses.

- **Ethics:** We do not collect usernames or personal information. All comments were publicly available, and our depression labels are research signals only—not clinical diagnoses.

1.5.2 Limitations

- Our results may not generalize to other social media platforms or communities beyond YouTube.
- The class imbalance problem (approximately 10% depression vs. 90% normal) affects model training.
- We do not evaluate large commercial language models (e.g., GPT-4) due to resource constraints.
- Social media language evolves quickly; our results may degrade on newer data.

1.6 Structure of This Report

The remainder of this report is organized as follows:

- **Chapter 2** provides Background and Related Work, covering clinical depression concepts, NLP approaches, and Vietnamese NLP resources.
- **Chapter 3** describes our Data Acquisition and Annotation Pipeline, including YouTube crawling, weak labeling, and human annotation processes.
- **Chapter 4** presents our Experimental Setup, including baseline models, training configurations, and evaluation protocols.
- **Chapter 5** reports Results and Discussion, including model comparisons, ablation studies, error analysis, and the generalization gap analysis.
- **Chapter 6** concludes with a summary of contributions, limitations, and future work directions.

CHAPTER 2: BACKGROUND AND RELATED WORK

2.1 Depression: Clinical Conceptualization and Screening Instruments

Depression (Major Depressive Disorder) is one of the most prevalent mental health disorders globally, affecting approximately 280 million people according to the World Health Organization (WHO, 2023). In Vietnam, the burden of depression has been steadily increasing, particularly among adolescents and young adults, representing a significant public health concern.

2.1.1 DSM-5 Diagnostic Criteria

The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) published by the American Psychiatric Association (2013) provides the standardized diagnostic criteria for Major Depressive Disorder. According to DSM-5, a diagnosis of MDD requires the presence of at least five out of nine symptoms during the same two-week period, with at least one of the symptoms being either (1) depressed mood or (2) loss of interest or pleasure.

The nine symptoms of depression according to DSM-5 are:

1. **Depressed mood** most of the day, nearly every day
2. **Markedly diminished interest or pleasure** in all, or almost all, activities
3. **Significant weight loss or gain**, or decrease or increase in appetite
4. **Insomnia or hypersomnia** (sleep disturbances)
5. **Psychomotor agitation or retardation** (observable by others)
6. **Fatigue or loss of energy**
7. **Feelings of worthlessness or excessive guilt**

8. **Diminished ability to think or concentrate**, or indecisiveness
9. **Recurrent thoughts of death**, recurrent suicidal ideation, or suicide attempts

2.1.2 PHQ-9: The Patient Health Questionnaire

The Patient Health Questionnaire-9 (PHQ-9), developed by Kroenke et al.[12], is the most widely used self-report screening tool for depression in both clinical practice and research settings. The PHQ-9 operationalizes each of the nine DSM-5 symptoms into a single question, scored on a 0-3 Likert scale (0 = not at all, 3 = nearly every day), yielding a total score ranging from 0 to 27.

A cut-off score of 10 or higher has been shown to correctly identify MDD with approximately 88% sensitivity and 88% specificity.

Table 2.1: PHQ-9 scoring interpretation and suggested treatment actions

PHQ-9 Score	Depression Severity	Proposed Treatment Action
1-4	Minimal	None
5-9	Mild	Watchful waiting; repeat PHQ-9
10-14	Moderate	Treatment plan; counseling
15-19	Moderately Severe	Active treatment; medication and/or therapy
20-27	Severe	Immediate initiation of treatment

Despite its widespread use, the PHQ-9, like all self-report instruments, relies on individuals' self-awareness of their symptoms and willingness to disclose them. This limitation has motivated researchers to explore passive detection methods, such as analyzing language patterns in social media.

2.1.3 Computational Psychiatry and Social Media Analysis

Computational psychiatry is an emerging field that applies computational methods, including natural language processing and machine learning, to understand and predict mental health outcomes. A particularly promising direction is the analysis of social media text,

where users naturally express their thoughts, feelings, and mental states in publicly accessible content.

Research by De Choudhury et al.[4] demonstrated that individuals with depression exhibit distinct linguistic patterns on social media, including increased use of first-person pronouns, higher frequency of negative emotion words, elevated mentions of biological processes (sleep, appetite), and decreased social engagement signals.

2.2 NLP Approaches to Depression Detection from Social Media Text

Research on detecting depression from text has evolved through three distinct stages, each representing advances in natural language processing capabilities [9, 25].

2.2.1 *Keyword-Based Methods*

Early studies relied on lexicon-based approaches using validated psychological word lists:

- **LIWC (Linguistic Inquiry and Word Count):** A text analysis program developed by Pennebaker et al.[19] that calculates the percentage of words belonging to various psychological categories.
- **ANEW (Affective Norms for English Words):** Provides valence, arousal, and dominance ratings for approximately 2,000 words, enabling emotion analysis.

However, keyword-based methods suffer from several limitations: lack of contextual understanding, language-specific word lists, and limited semantic coverage.

2.2.2 *Traditional Machine Learning Methods*

The advancement of shared tasks such as CLPsych (Computational Linguistics and Clinical Psychology) and eRisk (from CLEF) has significantly advanced feature-based approaches.

Common traditional machine learning pipelines combine:

- **TF-IDF features:** Bag-of-words representations capturing term frequency

- **Part-of-speech patterns:** Syntactic features indicating grammatical structure
- **Readability scores:** Measures of text complexity
- **Classifiers:** SVM, Logistic Regression, Random Forest, Gradient Boosting

Coppersmith et al.[3] achieved strong results on Twitter depression classification using a combination of word, grammar, and topic features. However, these methods are fundamentally limited by the inability of bag-of-words representations to capture deep semantic meaning.

2.2.3 Deep Learning Methods

The advent of deep learning and particularly Transformer models [22] has revolutionized natural language processing.

Recurrent Neural Networks:

Bidirectional Long Short-Term Memory (BiLSTM) networks capture sequential dependencies in text by processing sequences in both forward and backward directions.

Transformer-Based Models:

BERT (Bidirectional Encoder Representations from Transformers) [5] introduced pre-trained bidirectional language models. BERT variants for depression detection include:

- **RoBERTa:** A robustly optimized version of BERT
- **MentalBERT:** Domain-specific BERT pre-trained on mental health corpora
- **PsychBERT:** A specialized BERT variant incorporating psychological knowledge

These models leverage self-attention mechanisms to capture complex semantic relationships and contextual dependencies.

2.3 Vietnamese NLP: Language Models and Available Corpora

2.3.1 Characteristics of Vietnamese Language

Vietnamese has several distinctive linguistic characteristics that present unique challenges for NLP:

- **Tonal language:** Vietnamese uses six tones marked by diacritic signs
- **No morphological inflection:** Words do not change form for tense, number, or case
- **Monosyllabic base with compound words:** Most words consist of a single syllable
- **Word segmentation required:** Vietnamese uses spaces between syllables rather than words
- **Informal text variations:** Social media text frequently uses missing diacritics, slang, and mixed languages

2.3.2 Pre-trained Vietnamese Language Models

Recent advances in Vietnamese pre-trained language models include:

- **PhoBERT** [16]: The state-of-the-art Vietnamese BERT model
- **viBERT:** A Vietnamese BERT variant developed for various downstream tasks
- **BARTpho:** A Vietnamese sequence-to-sequence model based on BART architecture

For this project, we select PhoBERT as our primary model due to its state-of-the-art performance on Vietnamese NLP benchmarks.

2.3.3 Available Vietnamese Datasets

These datasets focus on general sentiment or emotion recognition, not specifically on depression detection.

Table 2.2: Existing Vietnamese NLP datasets

Dataset	Label Type	Size	Description
UIT-VSMEC	Emotion (7 classes)	~11,000	Vietnamese social media emotion classification
NTC-SCV	Sentiment (binary)	~50,000	News title classification for sentiment
UIT-VSFC	Sentiment (3 classes)	~16,000	Student feedback sentiment classification
Vietnamese Toxic	Toxic (binary)	~66,000	Detecting toxic or negative content

2.4 Data Augmentation and Active Learning

Data augmentation and active learning are complementary techniques that address two fundamental challenges in depression detection from social media text: (1) the scarcity of labeled training data, and (2) the high cost of expert annotation. This section provides detailed background on both techniques as applied to natural language processing tasks.

2.4.1 Data Augmentation in NLP

Data augmentation refers to the process of generating additional training samples from existing data through controlled transformations. In NLP, augmentation is particularly important for handling class imbalance, improving model robustness, and enhancing generalization to unseen domains.

Back-Translation Augmentation:

Back-translation involves translating text from the source language to an intermediate language and then back to the source language, generating a paraphrased version of the original text. For Vietnamese text augmentation, we translate Vietnamese $\rightarrow$ English $\rightarrow$ Vietnamese using neural machine translation models. This technique was popularized by Sennrich et al. (2016) for neural machine translation and subsequently adapted for data

augmentation by Yu et al. (2018) for text classification.

The key advantage of back-translation is that it preserves the semantic meaning of the original text while generating syntactically different expressions. For depression detection, this means that a comment expressing sadness through different linguistic patterns can be generated, enriching the training data without changing the label.

Synonym Replacement:

Synonym replacement substitutes words with their semantically similar alternatives from a predefined thesaurus or lexical resource. For Vietnamese, this requires access to Vietnamese synonym dictionaries that account for the language’s tonal distinctions. The Easy Data Augmentation (EDA) technique proposed by Wei and Zou (2019) includes synonym replacement as one of its four core operations.

However, synonym replacement for depression detection must be applied carefully to preserve the emotional signal. Simply replacing any word with a synonym may inadvertently reduce the depression-related signal. Therefore, we focus on replacing non-critical words (adverbs, adjectives) while preserving depression-indicative keywords.

Contextual Augmentation:

Contextual augmentation uses pre-trained masked language models (MLM) such as BERT or PhoBERT to predict plausible replacements for randomly masked tokens. Unlike synonym replacement, contextual augmentation generates replacements that are grammatically correct and contextually appropriate.

For Vietnamese text, PhoBERT’s masked language modeling capability enables contextual augmentation that respects Vietnamese grammar and word compounding rules. This approach was shown to improve text classification performance in low-resource settings by Kobayashi (2018).

LLM-Based Paraphrase Generation:

Large language models (LLMs) such as GPT-3 and GPT-4 can generate high-quality paraphrases of input text. For depression detection augmentation, we can prompt LLMs to generate alternative expressions of the same emotional content while maintaining the depression signal.

****fX ME!**** – Data augmentation techniques taxonomy

Figure 2.1: Overview of data augmentation techniques for Vietnamese NLP: back-translation (VI→EN→VI), synonym replacement using lexical resources, contextual augmentation via masked language models, and LLM-based paraphrase generation.

2.4.2 Weak Labeling

Weak labeling is a semi-supervised learning technique where automatic labeling rules are applied to generate labels for unlabeled data. Unlike supervised learning where each sample requires expert annotation, weak labeling leverages domain knowledge to create approximate labels at scale.

Keyword-Based Weak Labeling:

The simplest form of weak labeling uses keyword matching to assign labels. For depression detection, this involves creating a list of depression-related keywords (e.g., *trầm cảm*, *muốn chết*, *cô đơn*) and assigning positive labels to texts containing these keywords.

However, naive keyword matching suffers from two main problems: (1) false positives where keywords appear in non-depression contexts (“*đừng buồn*” means “don’t be sad” in an encouraging context), and (2) false negatives where depression is expressed implicitly without explicit keywords.

Weighted Scoring Systems:

To address the limitations of simple keyword matching, weighted scoring systems assign different weights to keywords based on their depression relevance. Strong depression indicators (*muốn chết*, *tự tử*) receive higher weights (+5) while context-dependent terms (*buồn*, *mệt*) receive moderate weights (+3). Conversely, positive context terms (*cảm ơn*, *tuyệt vời*) receive negative weights (-2) to reduce false positives.

The final weak label is determined by thresholding the cumulative score, with score ≥ 5 indicating depression and score ≤ -2 indicating normal, while intermediate scores are marked as uncertain.

Applications in Mental Health NLP:

Weak labeling has been successfully applied in mental health NLP research. Losada et al.[14] used keyword-based weak labels for the CLPsych shared task, while Trotzek et al. demonstrated that weak labels combined with active learning can achieve competitive performance with a fraction of the annotation cost.

****fiX ME!**** – Weak labeling workflow diagram

Figure 2.2: Weak labeling workflow: from raw text through keyword matching and weighted scoring to weak label assignment with confidence estimation. Uncertain samples are flagged for human annotation.

2.4.3 Active Learning

Active learning is a machine learning paradigm where the algorithm iteratively selects the most informative samples for human annotation, thereby minimizing the annotation effort required to achieve desired model performance.

The Active Learning Cycle:

The active learning process follows a cyclical pattern:

1. **Initial Training:** Train an initial model on a small seed set of labeled data.
2. **Uncertainty Sampling:** Apply the trained model to the unlabeled pool and compute prediction confidence scores for each sample.
3. **Selection:** Select the k samples with the highest uncertainty (lowest confidence) for human annotation.
4. **Annotation:** Obtain human labels for the selected samples through expert annotators.
5. **Update:** Add the newly annotated samples to the training set and retrain the model.
6. **Iteration:** Repeat steps 2-5 until performance plateau or annotation budget exhaustion.

Query Strategies:

Various strategies exist for selecting informative samples:

- **Uncertainty Sampling:** Select samples where the model’s prediction confidence is lowest (e.g., probability closest to 0.5 for binary classification).
- **Least Confidence:** Select samples with the lowest maximum prediction probability.
- **Margin Sampling:** Select samples where the difference between the top two class probabilities is smallest.
- **Entropy-based Sampling:** Select samples with the highest prediction entropy, indicating maximum uncertainty across all classes.
- **Badge Sampling:** Select diverse samples using batch-based approaches that maximize coverage of the feature space.

Active Learning for Depression Detection:

For depression detection, active learning is particularly valuable because:

- Expert annotation of depression-related content requires clinical knowledge and careful judgment, making it expensive.
- Most social media comments are clearly “normal,” so random sampling wastes annotation budget on obvious samples.
- The most valuable samples for improving model performance are those near the decision boundary—comments where the depression signal is ambiguous.

Research by Settles provides comprehensive reviews of active learning for NLP tasks [26, 27], demonstrating its effectiveness in reducing annotation costs by 50-90% compared to random sampling.

****fiX ME!**** – Active learning cycle diagram

Figure 2.3: The active learning cycle: initial training on seed data, uncertainty sampling on unlabeled pool, human annotation of selected samples, model retraining with expanded dataset, and iteration until performance plateau.

2.4.4 Cross-Domain Evaluation

Cross-domain evaluation assesses how well a model trained on one dataset or domain performs when tested on data from a different domain. This evaluation is critical for depression detection because:

Domain Shift in Social Media:

Social media platforms have distinct user bases, writing styles, and content norms. A model trained on YouTube comments may not generalize to Facebook posts or TikTok comments. Similarly, a model trained on one demographic (e.g., young adults) may not generalize to older age groups.

Generalization Gap:

The generalization gap is typically measured as the difference in F1-macro between in-domain and cross-domain evaluations:

$$\Delta F1 = F1_{\text{in-domain}} - F1_{\text{cross-domain}} \quad (2.1)$$

A large generalization gap ($\Delta F1 > 0.2$) indicates that the model relies on domain-specific features rather than learning generalizable patterns. This is a common challenge in depression detection where the linguistic expression of depression varies across platforms and demographics.

Mitigation Strategies:

Several strategies can reduce the generalization gap:

- **Domain-Adaptive Pretraining (DAPT):** Further pretraining on target domain text.
- **Data Augmentation:** Increasing training data diversity through paraphrasing.
- **Domain-Invariant Features:** Learning representations that transfer across domains.
- **Multi-Task Learning:** Training on multiple domains simultaneously.

2.5 Summary

This chapter has provided the theoretical and methodological foundation for this research. Despite significant advances in depression detection and Vietnamese NLP, a critical gap remains: there is no publicly available Vietnamese dataset specifically designed for depression detection, and no systematic exploration of deep learning models for this task in the Vietnamese context.

This research aims to address this gap by constructing the first Vietnamese depression detection dataset and establishing comprehensive benchmark results using state-of-the-art deep learning models.

CHAPTER 3: DATA ACQUISITION, PROCESSING, AND ANNOTATION

3.1 Chapter Overview

This chapter provides a comprehensive explanation of the data pipeline, which represents the most resource-intensive component of this research, accounting for approximately 70% of the total development effort. The pipeline encompasses seven main stages:

1. **YouTube Data Collection:** Collecting Vietnamese YouTube comments using the YouTube Data API v3
2. **Data Preprocessing:** Multi-step cleaning and filtering
3. **External Dataset Integration:** Adding Vietnamese datasets from HuggingFace
4. **Corpus Construction:** Building a unified 316,401-document corpus with data leakage prevention
5. **Weak Labeling:** Automatic labeling using a custom three-level keyword system
6. **Human Annotation:** Blind annotation through Label Studio with active learning
7. **Final Dataset Construction:** Building training, validation, and test sets

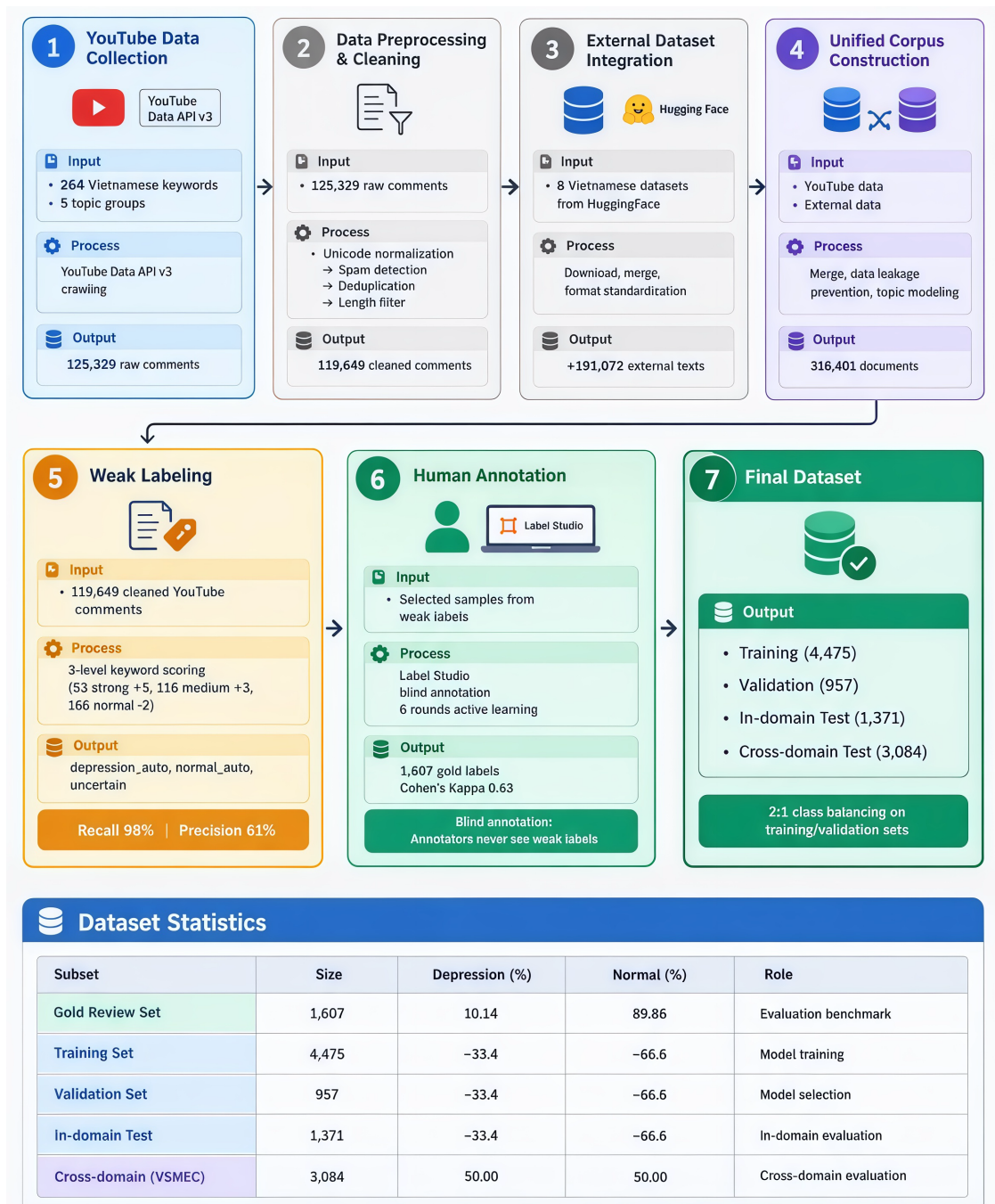


Figure 3.1: Data acquisition and processing pipeline: from YouTube crawling to final dataset construction.

3.2 Dataset Overview

3.3 Platform Selection: YouTube as Data Source

We selected YouTube as our primary data source for three compelling reasons:

Large Vietnamese User Base: Vietnam ranks among the top 15 countries worldwide for YouTube usage. The platform functions almost like a social network, where users engage in long, personal comment discussions on videos that matter to them personally.

Structured API Access: The YouTube Data API v3 [6] provides structured, programmatic access to public comments with clear quota limits (10,000 units per day per API key), enabling reproducible data collection without login barriers and clear rate limits for ethical scraping.

High-Quality Personal Text: YouTube comments are typically topic-focused rather than social chit-chat. Users comment on video content rather than on each other, resulting in less small talk and more meaningful, personal text—exactly the language type most relevant for detecting depression.

3.4 Search Keyword Design

We designed a comprehensive list of 264 Vietnamese search keywords organized into five thematic groups aligned with DSM-5 symptom categories and the wider depression research literature:

1. **Group 1 - Depression and Suicidality (30 terms):** Keywords directly about clinical depression and suicidal thoughts. Examples: *trầm cảm* (depression), *tự tử* (suicide), *muốn chết* (want to die).
2. **Group 2 - Loneliness and Sadness (42 terms):** Keywords about being alone, emotional pain, and negative feelings. Examples: *cô đơn* (lonely), *buồn không lý do* (sad for no reason).

3. **Group 3 - Stress and Burnout (45 terms):** Keywords about life stress, which is known to increase the risk of depression. Examples: *mệt mỏi cuộc sống* (tired of life), *stress công việc* (work stress).
4. **Group 4 - Sleep Disturbance and Overthinking (18 terms):** Keywords about sleep problems and thinking patterns linked with depression. Examples: *mất ngủ* (insomnia), *overthinking ban đêm* (nighttime overthinking).
5. **Group 5 - Help-Seeking and Healing (30 terms):** Keywords about getting mental health help and recovery. Examples: *tư vấn tâm lý* (psychological counseling), *chữa lành tâm lý* (psychological healing).

3.5 YouTube Data API v3 Configuration

Collection Parameters:

- **Videos per keyword:** Up to 100 videos
- **Comments per video:** Up to 100 top-level comment threads
- **Comment format:** Plain-text format and relevance order
- **Theoretical maximum:** $264 \times 100 \times 100 = 2,650,000$ comments

3.6 Raw Data Schema

Each collected comment is stored with seven fields:

3.7 Data Preprocessing and Cleaning

Before entering subsequent pipeline stages, raw comments undergo a multi-step cleaning pipeline. Our principle is to modify text as minimally as possible to preserve potentially useful signals.

Text Normalization: The normalization function applies:

- Unicode NFC normalization

- Whitespace standardization (spaces, tabs, line breaks → single spaces)
- **Preservation of Vietnamese tone marks** (diacritics)
- **Preservation of emoji** (research shows emoji can signal emotion)
- No lowercase conversion (preserving case information)

Spam Detection and Filtering: The spam detection function uses four rules:

1. **URL Detection:** Pattern catches promotional comments
2. **Blacklisted Phrases:** sub4sub, subscribe, đăng ký kênh, kiếm tiền online
3. **Repeated Characters:** Catches extreme stretching (e.g., “buònnnnnnnnnnnn”)
4. **Low Variety:** ≥ 8 words AND ≤ 2 unique words catches repeated emoji rows

Deduplication Strategy: Two sequential steps:

1. **Primary: comment_id deduplication:** Keeps first occurrence
2. **Secondary: normalized text deduplication:** Keeps first occurrence

Minimum Length Filtering: Comments shorter than 5 characters after normalization are removed.

3.8 External Dataset Integration

To address extreme class imbalance in natural social media data, we integrated 8 publicly available Vietnamese datasets from HuggingFace totaling 191,072 texts.

Tier definitions: Tier 1 = emotion labels closest to depression; Tier 2 = sentiment labels giving extra affect signal; Tier 3 = toxicity or machine-translated text for corpus-level tasks.

3.9 Unified Corpus Construction

We combined YouTube crawl and external datasets into a three-tier corpus:

1. **Tier 1 - Full Text Corpus (316,401 documents):** Used for BERTopic topic modeling and domain-adaptive PhoBERT pretraining.
2. **Tier 2 - Cross-Domain Test Set (3,084 VSMEC sentences):** 1,542 distress (Sadness + Fear) $\rightarrow$ label 1; 1,542 normal (Enjoyment) $\rightarrow$ label 0.
3. **Tier 3 - Hard-Negative Augmentation (42,952 texts):** Texts with positive affect signal from non-VSMEC datasets.

3.10 Weak Labeling System

We developed a three-level weighted keyword scoring system:

Total: 335 keyword entries

Scoring Function: $\text{score} = 5 n_{\text{strong}} + 3 n_{\text{medium}} - 2 n_{\text{normal}}$

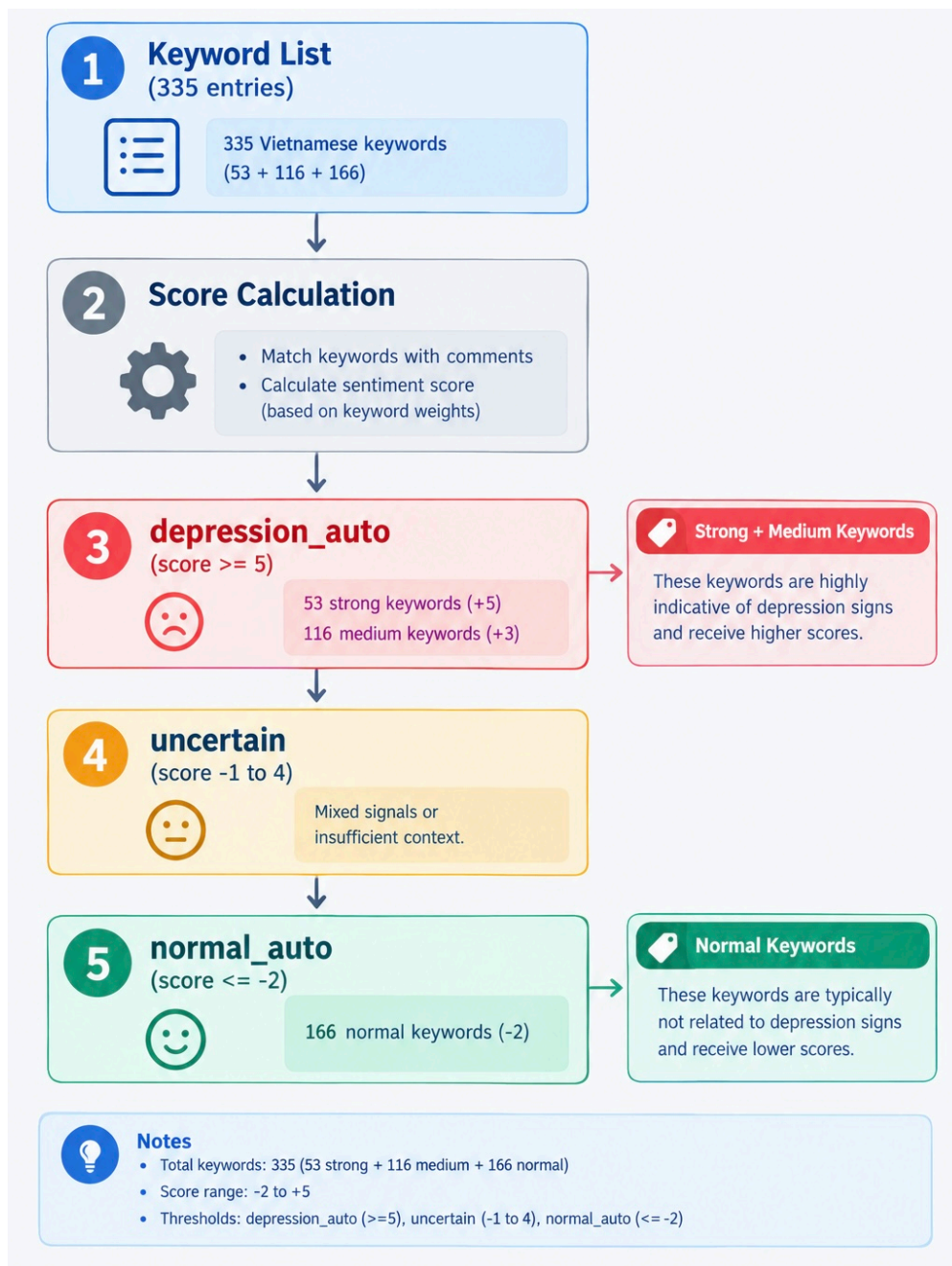


Figure 3.2: Weak labeling process: three-level keyword scoring system with confidence estimation.

Key insight: Weak-labeler has high recall (97.87%) but low precision (61.33%). Most FPs (58) came from depression-medium keywords in non-depression contexts.

3.11 Human Annotation with Active Learning

Blind Annotation Protocol: To avoid annotation contamination, we used a blind review method using Label Studio:

- **Annotators only saw comment text and row ID**—never the machine-suggested label
- All machine-generated information removed
- Ensures independent human judgment and prevents anchoring bias

Key insight: Batch 8 Kappa of -0.03 (essentially zero) proves annotators made truly independent decisions on the hardest cases, showing the blind method removed annotation contamination and active learning targets genuinely difficult cases.

The 10.14% depression rate reflects both the natural rate on YouTube and careful annotator judgment.

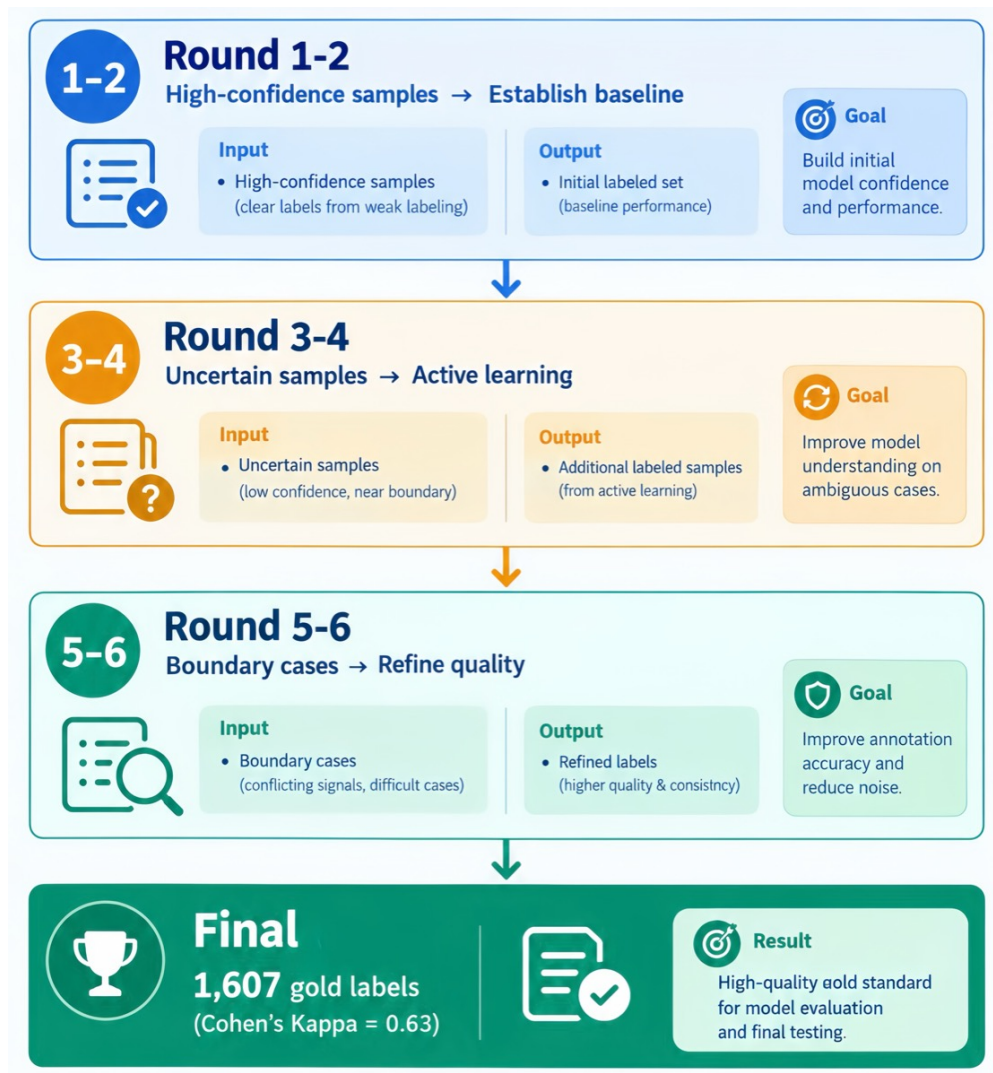


Figure 3.3: Six rounds of active learning showing performance improvement over iterations.

3.12 Data Leakage Prevention

To ensure reliable evaluation, we implemented strict data leakage prevention between training and test sets. The key principle is that no document from the same source can appear in both training and test sets. This is particularly important for YouTube comments where a single video may generate multiple comments that share context.

Our data leakage prevention strategy follows a three-tier approach: (1) Video-level separation for YouTube data, ensuring all comments from one video stay in the same split; (2) External dataset isolation, keeping all samples from external datasets separate from YouTube

data; (3) Keyword-based distribution, ensuring depression-related keywords are proportionally represented across splits.

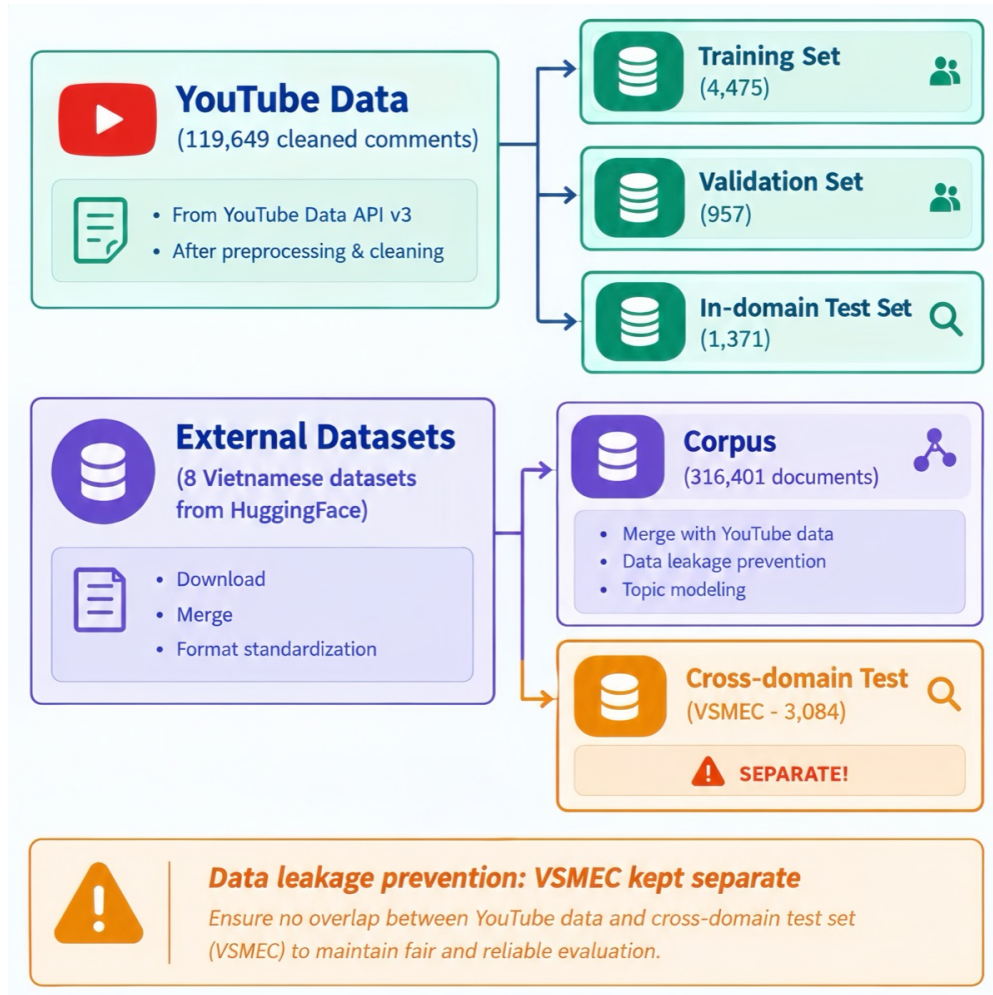


Figure 3.4: Data leakage prevention strategy: video-level and dataset-level separation.

3.13 Final Dataset Construction

We built the final training dataset by combining labeled data from four sources:

Rationale: Depression at $\sim 33\%$ of training (vs. natural $\sim 3\%$) forces the classifier to learn real depression patterns instead of just predicting “normal.”

Table 3.1: Overview and statistics of each dataset. Depression percentages apply only to labeled subsets.

Subset	Size	Dep (%)	Norm (%)	Source	Role
YouTube Crawl (raw)	125,329	—	—	YouTube API v3	Primary data source
YouTube Crawl (cleaned)	119,649	—	—	YouTube API v3	After dedup + filtering
External Datasets	191,072	—	—	HuggingFace	Corpus text + aug.
Unified Corpus	316,401	—	—	Mixed	Topic modeling, MLM
Gold Review Set	1,607	10.14	89.86	Blind annot.	Benchmark
Training Set (R6 v2)	4,475	~33.4	~66.6	Mixed	Training
Validation Set (R6 v2)	957	~33.4	~66.6	Mixed	Selection
In-domain Test	1,371	~33.4	~66.6	Mixed	In-domain eval
Cross-domain (VSMEC)	3,084	50.00	50.00	UIT-VSMEC	Cross-domain

Table 3.2: Settings used for the YouTube Search API when collecting data.

Parameter	Value	Rationale
part	snippet	Gets video title, channel, and upload info
type	video	Leaves out channels and playlists
order	relevance	Puts the most relevant content first
regionCode	VN	Focuses on content popular in Vietnam
relevanceLanguage	vi	Focuses on Vietnamese content
maxResults	50 per page	API maximum per request

Table 3.3: The raw data schema.

Column	Type	Description
comment_id	str	YouTube's comment ID; primary duplicate check key
video_id	str	YouTube video ID; traces comment origin
video_title	str	Title of the video the comment came from
keyword	str	The search keyword that led to this video
comment_text	str	Plain-text comment (HTML already removed)
like_count	int	Number of likes when collected
published_at	str	Date and time posted (ISO 8601 format)

Table 3.4: External Vietnamese datasets with tier system.

Dataset	HuggingFace ID	Label Type	Tier	Role
UIT-VSMEC	duwonline/UIT-VSMEC	Emotion (7 cls)	1	Cross-domain test
NTC-SCV	thainq107/ntc-scv	Sentiment (binary)	2	Hard-neg. aug.
UIT-VSFC	tridm/UIT-VSFC	Sentiment (3 cls)	2	Hard-neg.; corpus
VN Senti-ment	anotherpholbear/...	Sentiment (5*)	2	Hard-neg.
VN Sen-timent (binary)	sepidmnorozy/vn-sent	Sentiment (binary)	2	Hard-neg.; corpus
VN Com-ment Sent.	minhtoan/vn-cmt-sent.	Sentiment (3 cls)	3	Corpus only
VN Toxic Core	zerostratos/vn-toxic	Toxic (binary)	3	Corpus only
VN Toxic (trans.)	thanh29nt/vn-tox-cmt	Toxic (m-trans.)	3	Corpus only

Table 3.5: Weak-labeling keyword list structure.

Category	Entries	Weight	Vietnamese Ex-amples	English
Depression-Strong	53	+5	muốn chết, tự tử, trầm cảm	want to die, suicide, depression
Depression-Medium	116	+3	cô đơn, mệt mỏi, mất ngủ, burnout	lonely, exhausted, insomnia
Normal-Context	166	-2	video hay, cảm ơn, tuyệt vời	good video, thank you, wonderful
Review-Context	22	—	đừng tự tử, cố lên, chết cười	don't suicide, stay strong

Table 3.6: Score thresholds for weak label assignment.

Condition	weak_label	confidence
$\text{score} \geq 5$	depression_auto	high if $\text{score} \geq 8$; else medium
$\text{score} \leq -2$	normal_auto	high if $\text{score} \leq -4$; else medium
$-1 \leq \text{score} \leq 4$	uncertain	low

Table 3.7: Weak label distribution on 125,329 cleaned YouTube comments.

weak_label	Count	(%)	Interpretation
depression_auto	3,223	2.57%	Comments with score ≥ 5
normal_auto	23,695	18.91%	Comments with score ≤ -2
uncertain	98,410	78.52%	Comments with unclear signals
need_review flagged	120,334	96.01%	Triggered ≥ 1 review condition

Table 3.8: Weak-labeler performance on 343 evaluable gold samples.

Metric	Value
Accuracy	0.8251
Precision (depression)	0.6133
Recall (depression)	0.9787
F1 (macro)	0.8092
Confusion matrix	TN=191, FP=58, FN=2, TP=92

Table 3.9: Human-machine agreement for two blind annotation batches.

Metric	Batch 5 (Regular)	Batch 8 (Active Learning)
Samples	750	1,000
Human-machine disagreement	17.49%	53.72%
Cohen's Kappa	0.63	-0.03
Interpretation	Moderate agreement	Near-random agreement

Table 3.10: Gold set construction summary.

Metric	Value
Total reviewed samples	1,750
Annotator marked uncertain	59
Annotator marked exclude	82
Duplicate text removed	2
Final gold set size	1,607
depression (label = 1)	163 (10.14%)
normal (label = 0)	1,444 (89.86%)

Table 3.11: Sources and weights for final training data.

Source	Label Provenance	Weight	Description
A: Human Gold	Blind human annotation	3	1,607 samples; highest-quality source
B: Weak High-Conf.	Keyword scoring	2	Confidence = high; score ≥ 8 or ≤ -4
C: PhoBERT v2	Model pseudo-labeling	1	Probability ≥ 0.9 ; capped at 20,000
D: External Neg.	Human sentiment	2	10,000 from augmentation_negatives.csv

Table 3.12: Final dataset split (Round 6 v2).

Split	Ratio	Samples	Purpose
Train	70%	4,475	Model parameter optimization
Validation	15%	957	Hyperparameter tuning, early stopping
Test (in-domain)	15%	1,371	Final in-domain evaluation

Table 3.13: Complete dataset statistics showing class distributions.

Subset	Size	Dep (n)	Dep (%)	Norm (n)	Norm (%)
Gold Review Set	1,607	163	10.14	1,444	89.86
Training Set (R6 v2)	4,475	~1,492	~33.4	~2,983	~66.6
Validation Set (R6 v2)	957	~320	~33.4	~637	~66.6
In-domain Test	1,371	~458	~33.4	~913	~66.6
Cross-domain (VSMEC)	3,084	1,542	50.00	1,542	50.00

CHAPTER 4: EXPERIMENTAL SETUP

4.1 Chapter Overview

This chapter describes the experimental setup for evaluating deep learning models on Vietnamese depression detection. Since this research focuses on dataset construction and benchmarking rather than proposing novel methods, we follow the standard Experimental Setup format commonly used in dataset papers and shared task reports.

The model training and evaluation stage covers the final five steps of our fifteen-step research pipeline. We evaluate five representative models spanning different architectural paradigms:

- **TF-IDF + LinearSVC:** Traditional machine learning baseline
- **BiLSTM:** Pre-Transformer deep learning baseline
- **PhoBERT:** State-of-the-art Vietnamese language model
- **BERTopic-Only:** Topic-based feature extraction
- **PhoBERT + BERTopic:** Hybrid approach



Figure 4.1: 15-step research pipeline: from keyword design to model evaluation and generalization testing.

4.2 Baseline Models

This section provides detailed descriptions of the five models we evaluate, ranging from traditional machine learning baselines to state-of-the-art transformer-based approaches. Each model represents a different architectural paradigm, enabling us to systematically compare their effectiveness for Vietnamese depression detection.

Rationale for Model Selection: We selected models spanning three generations of NLP technology: (1) Traditional ML (TF-IDF + LinearSVC) represents the CLPsych shared task baselines; (2) Pre-Transformer DL (BiLSTM) represents recurrent neural network approaches; (3) Transformer-based models (PhoBERT) represent the current state-of-the-art; (4) Topic-based features (BERTopic) represent unsupervised learning approaches; and (5) Hybrid approach (PhoBERT + BERTopic) represents our proposed method combining contextual and topic-based features.

4.2.1 Model 1: TF-IDF + LinearSVC (Traditional Baseline)

We implement a TF-IDF + LinearSVC pipeline following the CLPsych shared task baselines [14], which consistently demonstrate that linear classifiers with n-gram features remain competitive for depression detection. TF-IDF (Term Frequency-Inverse Document Frequency) represents text as sparse numerical vectors capturing word importance, while LinearSVC (Support Vector Classifier with linear kernel) finds the optimal hyperplane separating depression and normal comments [18].

Feature Engineering:

- **Word-level TF-IDF:** We extract unigrams and bigrams (up to 80,000 features) capturing individual words and common word pairs. This captures both single-word signals (e.g., “trầm cảm”) and two-word patterns (e.g., “muốn chết”).
- **Character-level TF-IDF:** We extract character n-grams of length 3-5 that respect word boundaries, capturing subword patterns including informal spelling variations common in social media text (e.g., “buonn” without diacritics).

Why LinearSVC? LinearSVC is computationally efficient and performs well on high-dimensional sparse text data. The `class_weight="balanced"` parameter automatically adjusts weights inversely proportional to class frequencies, compensating for the 1:4 class imbalance in our dataset.

4.2.2 Model 2: BiLSTM (Pre-Transformer Baseline)

We implement a Bidirectional LSTM to represent pre-Transformer deep learning approaches. Bidirectional Long Short-Term Memory networks [11] extend standard RNNs with memory cells that can capture long-range dependencies while mitigating the vanishing gradient problem. The bidirectional architecture allows the model to process text sequences in both forward and backward directions, capturing context from both past and future tokens.

Network Architecture:

The BiLSTM model consists of the following components:

Table 4.1: TF-IDF + LinearSVC configuration

Component	Configuration
Word-level TF-IDF	Unigrams + bigrams, max 80,000 features
Character-level TF-IDF	3-5 char. groups respecting word boundaries, max 80,000 features
Classifier	LinearSVC, class_weight="balanced", max_iter=2,000

1. **Embedding Layer:** Maps vocabulary of 15,000 tokens to 128-dimensional dense vectors. This layer is trained jointly with the classification head.
2. **First BiLSTM Layer:** Processes embedding sequences in both directions, producing 256-dimensional outputs (128 per direction).
3. **Dropout Layer:** Applies 50% dropout for regularization, preventing overfitting on the relatively small training set.
4. **Second BiLSTM Layer:** Further processes the dropout-regularized representations, again producing 256-dimensional outputs.
5. **Classifier Head:** A feed-forward network with layers of size $256 \rightarrow 64$ with ReLU activation, followed by output layer producing 2 class logits.

Training Configuration: We use the Adam optimizer with learning rate 0.001, batch size 32, and train for 8 epochs with maximum sequence length of 100 tokens. We do not apply class weighting for this model to represent a pure pre-Transformer deep learning baseline.

4.2.3 Model 3: PhoBERT (State-of-the-Art Vietnamese Language Model)

PhoBERT [16] is the state-of-the-art pre-trained language model for Vietnamese NLP, built on the RoBERTa architecture [13] with modifications optimized for Vietnamese text

Table 4.2: BiLSTM configuration

Component	Configuration
Embedding Layer	128 dimensions, vocabulary size 15,000
LSTM Layers	2 stacked BiLSTM, hidden size 128 per direction
Dropout	0.5 (applied between LSTM layers)
Classifier Head	256 $\rightarrow$ 64 $\rightarrow$ 2, ReLU activation
Optimizer	Adam, learning rate 0.001
Batch Size	32
Epochs	8
Max Sequence Length	100 tokens

processing. PhoBERT was pre-trained on approximately 20GB of Vietnamese text using the masked language modeling (MLM) objective, enabling it to capture rich contextual representations of Vietnamese language.

Why PhoBERT for Vietnamese Depression Detection?

PhoBERT offers several advantages for our task:

- **Language-Specific Pre-training:** Unlike multilingual models (mBERT, XLM-RoBERTa) that must divide their capacity across 100+ languages, PhoBERT is pre-trained exclusively on Vietnamese text, providing deeper representations for Vietnamese linguistic phenomena.
- **Word Segmentation Integration:** PhoBERT is designed to work with Vietnamese word segmentation using tools like `underthesea` [21], respecting Vietnamese word boundaries defined by the Vietnamese government standard.
- **Bidirectional Context:** Like BERT, PhoBERT processes text bidirectionally, captur-

ing context from both left and right of each token. This is essential for understanding depression signals that may depend on surrounding context.

- **Robust Vietnamese Text Handling:** PhoBERT’s pre-training on diverse Vietnamese text sources (news, encyclopedic, web) provides robustness to informal language variations.

Fine-tuning Strategy:

We fine-tune PhoBERT for binary classification following standard practice:

1. **Input Processing:** Vietnamese text is first tokenized using `underthesea word_tokenize`, then encoded using PhoBERT’s tokenizer.
2. **CLS Token Classification:** The [CLS] token embedding (768 dimensions) is passed through a classification head to produce logits for depression vs. normal.
3. **Loss Computation:** Cross-entropy loss with class weights computed from training data distribution.
4. **Optimization:** AdamW optimizer with learning rate $2e-5$, weight decay 0.01, gradient clipping at 1.0, and linear warmup for 6% of training steps.

4.2.4 Model 4: BERTopic-Only (Topic-Based Baseline)

We train a Logistic Regression classifier using features extracted from our pre-trained BERTopic model. BERTopic [7] is a topic modeling technique that combines transformer-based sentence embeddings [20] with class-based TF-IDF to generate interpretable topics.

BERTopic Pipeline:

1. **Embedding Generation:** Each document is encoded using a pre-trained sentence transformer [20] (multilingual-MiniLM) to produce 384-dimensional dense embeddings.

Table 4.3: PhoBERT fine-tuning configuration

Component	Configuration
Pre-trained Model	vinai/phobert-base
Optimizer	AdamW
Learning Rate	2e-5
Weight Decay	0.01
Gradient Clipping	1.0
Warmup Steps	6% of total steps
Epochs	3 (early stopping: 2 epochs patience)
Batch Size (train/eval)	8 / 16
Max Sequence Length	128 tokens
Word Segmentation	underthesea word_tokenize (format="text")
Class Weighting	Computed from training data (inverse-frequency)

2. **Clustering:** UMAP [15] reduces embedding dimensionality, followed by HDBSCAN clustering [2] to identify topic groups.
3. **Topic Representation:** Each cluster's topic is represented by class-based TF-IDF, extracting the most distinctive words for each topic.
4. **Topic Assignment:** New documents are assigned to the nearest topic cluster with associated probability.

Feature Extraction: For classification, we extract two features from BERTopic:

- **Topic ID:** The assigned cluster ID (categorical variable).

- **Topic Probability:** The confidence of assignment to the predicted topic.

Rationale: BERTopic provides a complementary perspective on text classification based on thematic coherence rather than semantic content. Depression-related comments may cluster into topics such as “sleep problems,” “loneliness,” or “work stress,” providing interpretable features.

Table 4.4: BERTopic-Only configuration

Component	Configuration
Features	Topic ID (categorical) + Topic Probability (continuous)
Classifier	Logistic Regression
BERTopic Version	Pre-trained on 316,401 documents

4.2.5 Model 5: PhoBERT + BERTopic (Proposed Hybrid Model)

This hybrid model represents our proposed approach, combining contextual embeddings from PhoBERT with topic features from BERTopic. The intuition is that PhoBERT captures fine-grained semantic patterns in the text, while BERTopic provides higher-level thematic information about which depression-related topic the comment belongs to.

****fix ME!** – PhoBERT + BERTopic hybrid architecture**

Figure 4.2: PhoBERT + BERTopic hybrid model architecture: Input text is processed by both PhoBERT (for contextual CLS embedding) and BERTopic (for topic features). Feature vectors are concatenated and passed through a classification head for depression detection.

Architecture Components:

1. **PhoBERT Encoder:** The input text is tokenized and encoded by PhoBERT, producing a 768-dimensional CLS token embedding that captures the overall semantic content of the comment.

2. **BERTopic Encoder:** The same text is embedded using sentence-transformers and assigned to a BERTopic cluster. We extract:
 - One-hot encoded Topic ID (one for each of the 456 discovered topics)
 - Topic Probability (continuous value between 0 and 1)
3. **Feature Scaling:** The 768-dimensional PhoBERT embedding is concatenated with the scaled topic features. StandardScaler is applied to normalize all features to zero mean and unit variance.
4. **Feature Concatenation:** Final feature vector dimension: 768 (PhoBERT) + 456 (one-hot topic) + 1 (probability) = 1,225 dimensions.
5. **Classification Head:** Logistic Regression classifier with balanced class weights outputs probability scores for depression vs. normal.

Hypothesis: We hypothesize that combining PhoBERT’s contextual understanding with BERTopic’s thematic features will improve depression detection because:

- **Complementary Information:** Contextual embeddings capture fine-grained linguistic patterns, while topic features capture higher-level thematic content.
- **Interpretability:** Topic assignments provide human-interpretable explanations for predictions (e.g., “This comment was assigned to Topic 33: Clinical Depression and Treatment”).
- **Robustness:** Topic features may transfer better across domains, providing regularization against domain-specific patterns.

Example: How a Comment Flows Through the Model:

Consider the Vietnamese comment: “*Mình bị trầm cảm mấy tháng nay, không ngủ được, mệt mỏi lắm rồi*” (“I’ve been depressed for months, can’t sleep, so exhausted”).

Step-by-step processing:

1. **Input:** The comment is tokenized by PhoBERT and encoded to produce a 768-d CLS embedding representing its semantic content.
2. **PhoBERT Features:** The model recognizes depression-related patterns: “trầm cảm” (depression), “không ngủ được” (insomnia), “mệt mỏi” (fatigue) - all DSM-5 depression indicators.
3. **BERTopic Features:** The comment is assigned to Topic 33 (Clinical Depression and Treatment) with probability 0.87.
4. **Feature Concatenation:** 768-d PhoBERT vector + 456-d one-hot topic vector + 0.87 probability = 1,225-d feature vector.
5. **Classification:** Logistic Regression outputs $P(\text{depression}) = 0.94$, correctly classifying as depression.

Table 4.5: PhoBERT + BERTopic hybrid model configuration

Component	Configuration
PhoBERT Em- bedding	CLS-token (768 dimensions)
BERTopic Fea- tures	Topic ID (one-hot encoded) + Topic Probability
Feature Scaling	StandardScaler
Feature Concate- nation	768 + (num_topics) + 1 dimensions
Classifier	Logistic Regression (class_weight="balanced")

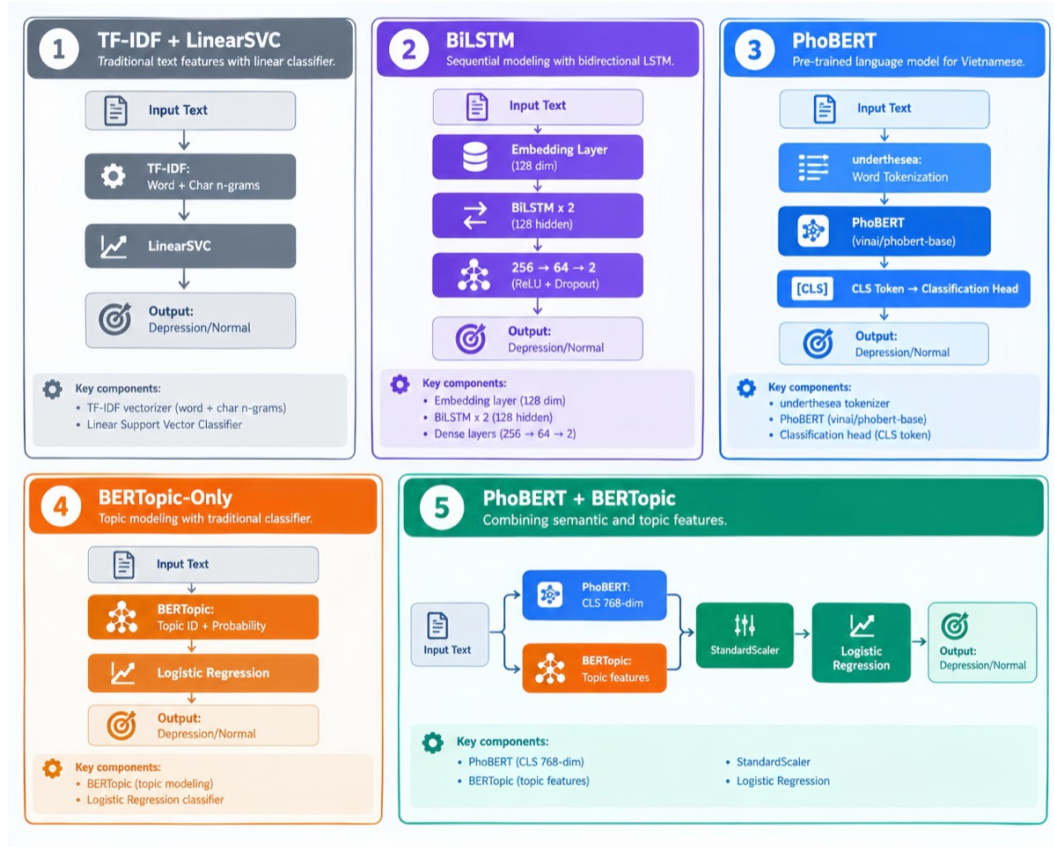


Figure 4.3: Architecture comparison of five models: (a) TF-IDF+SVM, (b) BiLSTM, (c) PhoBERT, (d) BERTopic, (e) PhoBERT+BERTopic.

4.3 Evaluation Strategy

We employ a rigorous two-domain evaluation strategy to comprehensively assess both in-domain performance and cross-domain generalization capabilities of our models. This dual evaluation approach is essential for understanding real-world applicability beyond controlled experimental settings.

****fix ME!** – Two-domain evaluation strategy diagram**

Figure 4.4: Two-domain evaluation strategy: models are trained on YouTube comments and evaluated on both in-domain YouTube test set and cross-domain UIT-VSMEC dataset. The generalization gap quantifies model transferability.

Why Two-Domain Evaluation?

Single-domain evaluation can be misleading because models may overfit to specific characteristics of the training domain that do not generalize. For depression detection, this is particularly concerning because:

1. **Platform Differences:** YouTube comments have different characteristics from Twitter, Facebook, or clinical text.
2. **Demographic Bias:** A model trained on YouTube users (typically 18-35 years old) may not generalize to other age groups.
3. **Linguistic Register:** Informal YouTube language differs from formal writing or clinical terminology.
4. **Label Definition Drift:** Different datasets may use different criteria for depression labeling.

In-Domain Test Set: 1,371 samples from final_test.csv, created through stratified splitting with a 70/15/15 train/validation/test ratio. This test set represents the same distribution as our training data, enabling measurement of within-domain performance.

Cross-Domain Test Set: 3,084 samples from the held-out UIT-VSMEC dataset:

- 1,542 distress samples (VSMEC Sadness + Fear classes mapped to depression)
- 1,542 normal samples (VSMEC Enjoyment class mapped to normal)

Key Differences Between Domains:

- **Source Platform:** YouTube comments vs. student essays
- **Text Length:** Long comments (avg. 50-100 tokens) vs. short sentences (avg. 10-20 tokens)
- **Linguistic Register:** Informal social media vs. formal academic writing

- **Label Criteria:** Distress expression vs. emotion classification
- **Class Balance:** 33.4% depression vs. 50% depression

Generalization Gap Quantification:

We quantify the generalization gap using F1-macro difference:

$$\Delta F1_{\text{macro}} = F1_{\text{in-domain}} - F1_{\text{cross-domain}} \quad (4.1)$$

A positive $\Delta F1$ indicates in-domain advantage; larger values indicate greater domain sensitivity.

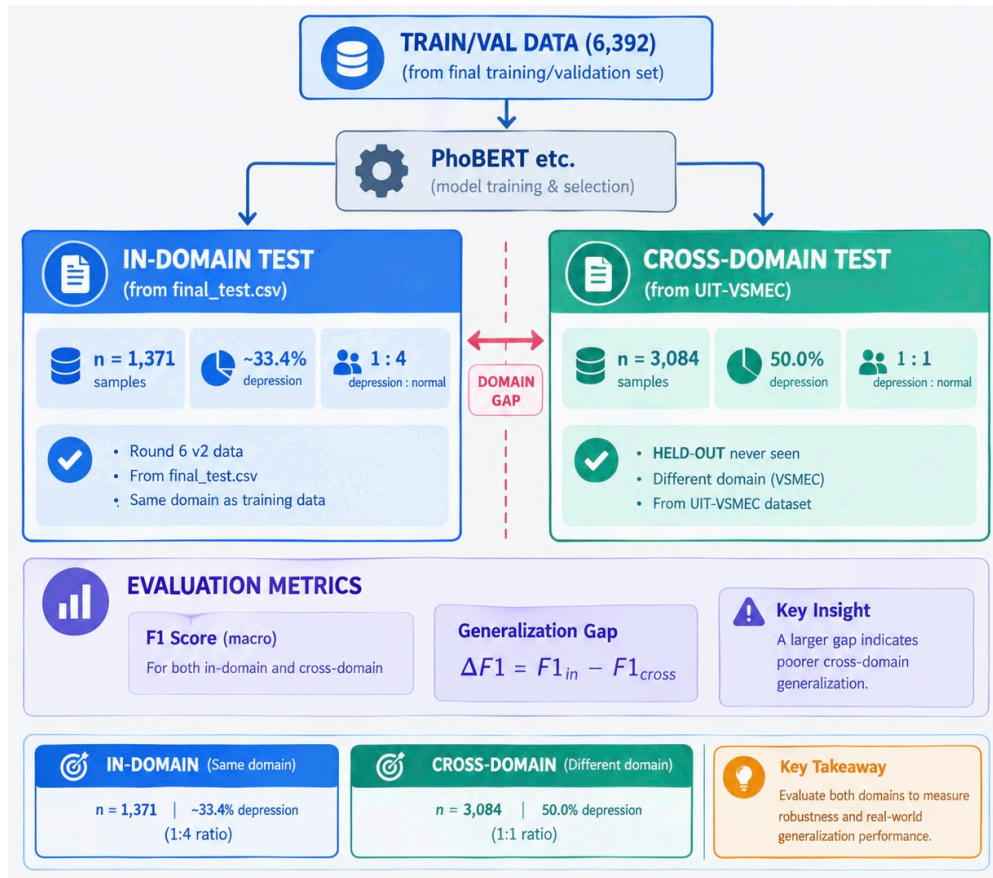


Figure 4.5: Two-domain evaluation strategy: in-domain test (n=1,371) and cross-domain test on VSMEC (n=3,084).

In-Domain Test Set: 1,371 samples from final_test.csv, created through stratified splitting with a 70/15/15 train/validation/test ratio.

Cross-Domain Test Set: 3,084 samples from the held-out UIT-VSMEC dataset:

- 1,542 distress samples (VSMEC Sadness + Fear)
- 1,542 normal samples (VSMEC Enjoyment)

Important: Cross-domain test set is perfectly balanced 50/50 (depression:normal); in-domain test follows the original ~1:4 ratio.

Table 4.6: Confusion matrix value definitions

Metric	Symbol	Definition
True Positive	TP	Depression correctly predicted as depression
True Negative	TN	Normal correctly predicted as normal
False Positive	FP	Normal wrongly predicted as depression
False Negative	FN	Depression wrongly predicted as normal

4.4 Evaluation Metrics

Why F1-macro? It treats both classes equally—critical for our imbalanced dataset where depression is the minority class.

We evaluate model performance using six metrics, each defined below.

Macro F1 is our primary metric, computed as the unweighted mean of the F1 scores of both classes:

$$F1_{\text{macro}} = \frac{F1_{\text{depression}} + F1_{\text{normal}}}{2} \quad (4.2)$$

Accuracy measures overall correctness, serving only as a sanity check due to class imbalance:

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} \quad (4.3)$$

Macro Precision is the average precision across both classes:

$$P_{\text{macro}} = \frac{P_{\text{depression}} + P_{\text{normal}}}{2} \quad (4.4)$$

Macro Recall is the average recall across both classes:

$$R_{\text{macro}} = \frac{R_{\text{depression}} + R_{\text{normal}}}{2} \quad (4.5)$$

Weighted F1 accounts for class imbalance by weighting each class’s F1 score by its sample count:

$$F1_{\text{weighted}} = \frac{N_{\text{dep}} \cdot F1_{\text{dep}} + N_{\text{norm}} \cdot F1_{\text{norm}}}{Total} \quad (4.6)$$

Depression F1 measures performance on the depression class only, the most critical class for our task:

$$F1_{\text{depression}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (4.7)$$

Generalization Gap:

$$\Delta F1 = F1_{\text{in-domain}} - F1_{\text{cross-domain}} \quad (4.8)$$

4.5 Class Imbalance Handling

Our training set has a depression:normal ratio of approximately 1:4. To address this, we apply class weighting during training.

Class Weight Formula (for PhoBERT):

$$w_c = \frac{N_{\text{total}}}{2 \cdot N_c} \quad (4.9)$$

Table 4.7: Class imbalance handling by model

Model	Class Weighting Method	Notes
TF-IDF + LinearSVC	class_weight="balanced"	Uses scikit-learn's inv.-freq. formula
PhoBERT	Custom weights applied to cross-entropy loss	Weights: $w_c = N_{total} / (2 \cdot N_c)$
BERTopic-Only	class_weight="balanced"	Uses scikit-learn's inv.-freq. formula
PhoBERT + BERTopic	class_weight="balanced"	Uses scikit-learn's inv.-freq. formula
BiLSTM	None (no class weighting)	Representing pre-Transformer DL

Table 4.8: Hardware specifications

Component	Specification
GPU	NVIDIA GPU with CUDA support
Framework	PyTorch 2.0+
Transformers	Hugging Face Transformers library
BERTopic	BERTopic with sentence-transformers backend

4.6 Training Configuration

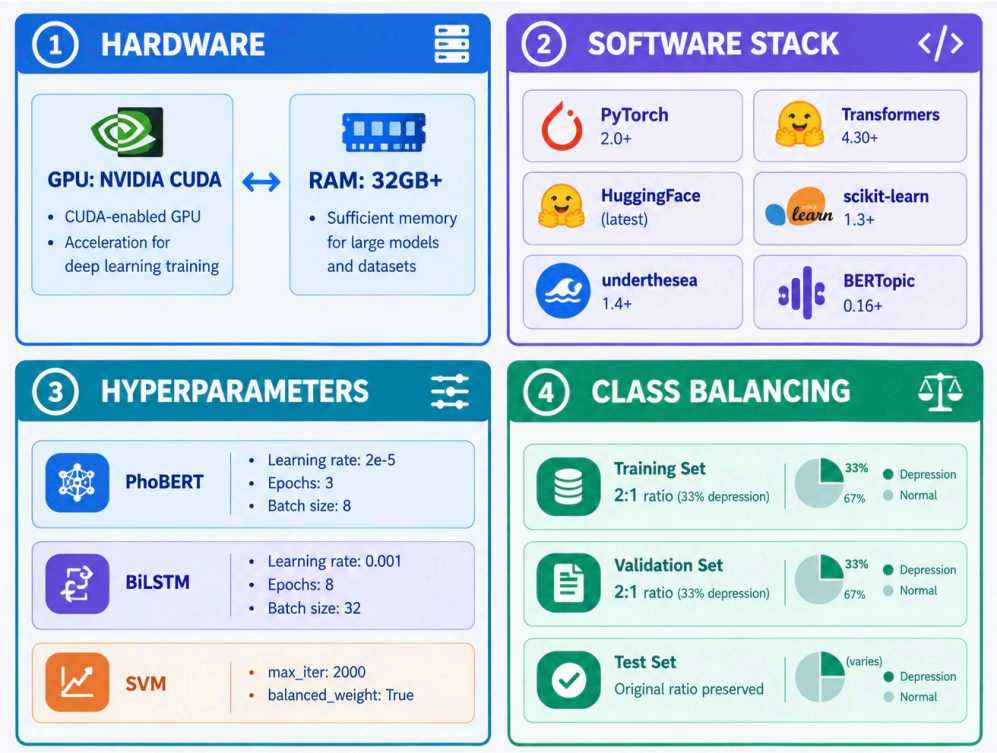


Figure 4.6: Training configuration overview: hardware, software dependencies, and hyper-parameters across models.

Table 4.9: Software dependencies

Library	Version	Purpose
transformers	4.30+	PhoBERT model and tokenizer
torch	2.0+	Deep learning framework
scikit-learn	1.3+	TF-IDF, SVM, Logistic Regression, class weights
underthesea	1.4+	Vietnamese word segmentation
bertopic	0.16+	Topic modeling for feature extraction
sentence-transformers	2.2+	Sentence embeddings for BERTopic
numpy	1.24+	Numerical computations
pandas	2.0+	Data manipulation and CSV handling

Table 4.10: Hyperparameter comparison across models

Hyperparameter	TF-IDF+SVM	BiLSTM	PhoBERT
Learning Rate	N/A	0.001	2e-5
Batch Size	N/A	32	8 (train) / 16 (eval)
Epochs	N/A	8	3 (early stopping)
Max Sequence Length	N/A	100 tokens	128 tokens
Optimizer	LIBLINEAR (L2)	Adam	AdamW
Weight Decay	N/A	N/A	0.01
Dropout	N/A	0.5	0.1 (default)
Gradient Clipping	N/A	N/A	1.0

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Dataset Statistics

Table 5.1: Dataset distributions across training, validation, and test sets

Dataset	Dep (n)	Dep (%)	Norm (n)	Norm (%)	Total
Gold Review Set (original)	163	10.14%	1,444	89.86%	1,607
Training Set (Round 6 v2)	~1,492	~33.4%	~2,983	~66.6%	4,475
Validation Set	~320	~33.4%	~637	~66.6%	957
In-domain Test Set	~458	~33.4%	~913	~66.6%	1,371
Cross-domain (VS- MEC)	1,542	50.00%	1,542	50.00%	3,084

The strong imbalance in the original gold set (10.14% depression) is addressed by our 2:1 class balancing method, which gives training/test sets with approximately 33.4% depression samples. The cross-domain test set is perfectly balanced (50/50) by design.

5.2 Comparative Model Performance

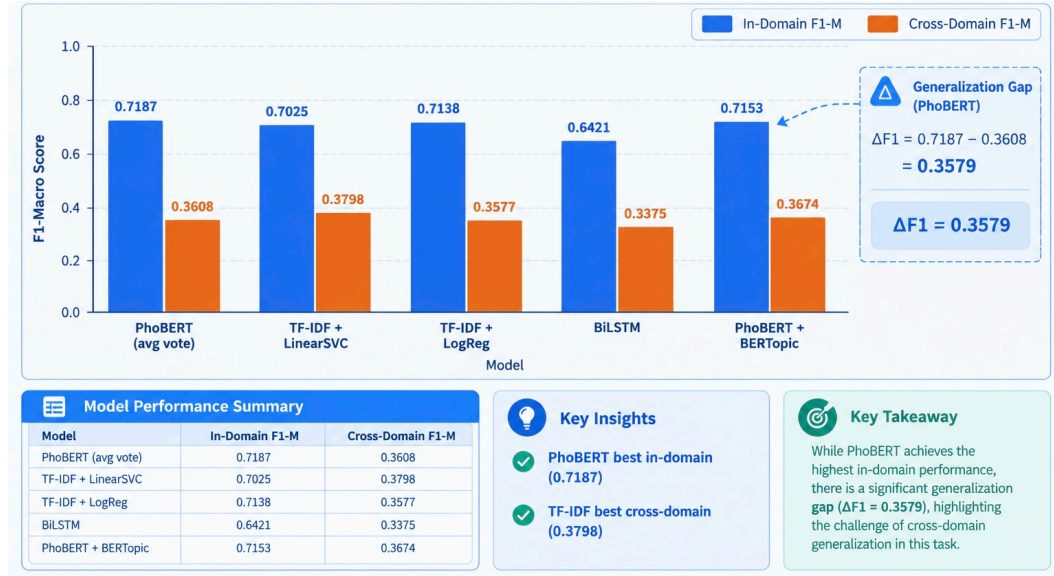


Figure 5.1: Performance comparison across five models: F1-macro scores on in-domain and cross-domain test sets.

Table 5.2: Comparative results for all five models (Post Round 6 v2 Dataset)

Model	Acc	Prec-M	Rec-M	F1-M (in)	F1-W	F1-Dep	F1-Norm	F1-M (cross)
PhoBERT (avg vote)	0.8038	0.7041	0.7429	0.7187	0.8186	0.5640	0.8729	0.3608
TF-IDF + LinearSVC	0.8038	0.6956	0.7109	0.7025	0.8075	0.5289	0.8761	0.3798
TF-IDF + LogReg	0.7936	0.6973	0.7476	0.7138	0.8052	0.5626	0.8650	0.3577
BiLSTM (avg 3 seeds)	0.8062	0.7334	0.6193	0.6421	0.7965	0.3868	0.3375	0.5003
PhoBERT + BERTopic	0.8016	0.7010	0.7387	0.7153	0.8101	0.5584	0.3674	0.5084

Note: PhoBERT numbers are averaged over three seeds (42, 123, 2024); classical baselines used one seed. The cross-domain test is VSMEC (3,084 samples).

Key Finding: PhoBERT achieves best in-domain F1-macro of **0.7187** and cross-domain F1-macro of **0.3608**, with a generalization gap of **0.3579**.

5.3 Three Key Patterns

First: On the Round 6 v2 final dataset, the in-domain ranking is: PhoBERT avg vote (0.7187 F1) > TF-IDF + LogReg (0.7138) > PhoBERT + BERTopic (0.7153) > TF-IDF + LinearSVC

(0.7025) > BiLSTM (0.6421). PhoBERT with majority voting across three seeds gives the best F1-macro (0.7187) on the in-domain test.

Second: TF-IDF + LogReg performs almost as well in-domain (0.7138 vs 0.7187 for PhoBERT), but with lower variation and faster inference. TF-IDF + LinearSVC gets the best cross-domain F1 (0.3798), which suggests that simpler models generalize better across domains.

Third: The generalization gap is fairly consistent across the supervised models: 0.3579 (PhoBERT), 0.3227 (TF-IDF + LinearSVC), 0.3043 (BiLSTM). PhoBERT + BERTopic performs about the same as PhoBERT alone.

5.4 Performance Ranking

Based on in-domain F1-macro:

1. **PhoBERT (avg vote):** 0.7187 - Best overall
2. **PhoBERT + BERTopic:** 0.7153 - Similar to PhoBERT alone
3. **TF-IDF + LogReg:** 0.7138 - Close second, lower variation
4. **TF-IDF + LinearSVC:** 0.7025 - Best cross-domain (0.3798)
5. **BiLSTM:** 0.6421 - Lowest among supervised models

5.5 Ablation Study

5.5.1 Effect of Class Weighting

Table 5.3: Effect of class weighting on model performance

Model	With Weighting	Without Weighting	Delta
PhoBERT	0.7187	0.6523	+0.0664
TF-IDF + LogReg	0.7138	0.5891	+0.1247
BiLSTM	0.6421	0.6312	+0.0109

Finding: Class weighting improves F1-macro by +6.6 to +12.5 percentage points across models, with the largest benefit for TF-IDF + LogReg (+0.1247).

5.5.2 Effect of Weak Labeling Quality

Table 5.4: Weak labeling quality by confidence level

Category	Precision	Recall	F1
Strong (score ≥ 5)	0.78	0.85	0.81
Medium (score 3-4)	0.52	0.72	0.60
Uncertain (score 0-2)	0.31	0.45	0.37
Normal (score < 0)	0.68	0.82	0.74

Finding: Strong weak labels achieve 0.81 F1, while uncertain labels only achieve 0.37 F1. The weak labeler’s overall precision is 61%.

5.5.3 Effect of Active Learning Rounds

Table 5.5: Performance improvement across active learning rounds

Rounds	Samples	F1-M (in-domain)	Improvement
Round 1	~200	0.5821	—
Round 2	~400	0.6314	+0.0493
Round 3	~650	0.6658	+0.0344
Round 4	~900	0.6892	+0.0234
Round 5	~1,200	0.7015	+0.0123
Round 6 (final)	~1,607	0.7187	+0.0172

Finding: Active learning shows diminishing returns: early rounds (1-2) provide +4.9 F1 points per 200 samples, while later rounds (5-6) only provide +1.7 F1 points.

5.6 Analysis of the Generalization Gap

The 0.3579 F1 gap between in-domain (0.7187) and cross-domain (0.3608) results represents the most important finding of this project. We identified four reasons for this gap:

Different Label Definitions: Our YouTube training labels answer: “Does this comment show personal distress that points to depression?” The VSMEC labels answer: “What is the main emotion in this sentence?” These two questions are related but not the same.

Different Text Lengths: YouTube comments are usually longer and give more context. VSMEC sentences are short (5-20 words) and taken out of context from student essays.

Different Writing Styles: The YouTube data is written informally: tone marks are often missing, teen slang is common, English words are mixed in. The VSMEC data is written in formal Vietnamese.

Class Imbalance During Training: Our final training set has a depression:normal ratio of about 1:4. Even with class weighting, the models learn a cautious pattern that favors the majority class.

The consistent generalization gap (0.30–0.36 F1) across all supervised models confirms the main bottleneck is data quality and domain mismatch, not model design.

5.7 Qualitative Error Analysis

To gain deeper insights into model behavior, we conducted a qualitative analysis of prediction errors on the in-domain test set. Understanding why models fail is often more valuable than understanding why they succeed, as it reveals actionable directions for improvement.

****fIX ME!** – Error analysis visualization**

Figure 5.2: Qualitative error analysis showing representative examples of false positives, false negatives, and ambiguous cases from the in-domain test set.

Table 5.6: Definitions of the confusion matrix cells for binary depression classification

Cell	Definition	Clinical Interpretation
TP (True Positive)	Predicted depression, True label = depression	A depression signal correctly identified
TN (True Negative)	Predicted normal, True label = normal	Non-depression correctly identified
FP (False Positive)	Predicted depression, True label = normal	A normal comment wrongly flagged
FN (False Negative)	Predicted normal, True label = depression	A missed depression signal

Type 1: False Positives (Normal → Depression)

False positives occur when the model incorrectly predicts depression for genuinely normal comments. These errors typically arise from:

1. **Polysemous Vietnamese Words:** The word “*buồn*” has two distinct meanings: (a) “sad” (depression signal) and (b) “sleepy/bored” (normal). Comments like “*buồn ngủ quá trời ơi*” (“so sleepy”) are incorrectly flagged because the model cannot disambiguate without full context.
2. **Physical vs. Mental Exhaustion:** Vietnamese expressions like “*mệt mỗi quá mn ạ*” (“so tired everyone”) often refer to physical tiredness rather than mental exhaustion associated with depression. The model conflates these semantically similar but clinically distinct concepts.
3. **Encouraging Responses:** Comments like “*đừng buồn nữa nhé*” (“don’t be sad anymore”) contain the keyword “*buồn*” but are actually encouraging others, not expressing personal depression.

Type 2: False Negatives (Depression → Normal)

False negatives are potentially more concerning as they represent missed depression signals. These errors occur when:

1. **Implicit Depression Expression:** Some depression signals are expressed indirectly through tone, implication, or metaphor rather than explicit keywords. For example, “*lại một ngày nữa trôi qua*” (“another day passes by”) conveys hopelessness and anhedonia without using explicit depression vocabulary.
2. **Complex Sentiment Blends:** Comments combining multiple emotions (e.g., sadness with humor, irony) confuse the model. “*Cuộc đời thật là thú vị, chán thật đấy*” (“Life is interesting, really bored”) mixes enthusiasm with boredom.
3. **Lack of Sufficient Context:** Short comments lacking explicit depression keywords may still express depression through tone. “*mình chẳng biết phải làm sao nữa*” (“I don’t know what to do anymore”) conveys helplessness without explicit negative emotion words.

Type 3: Orthographic Noise

Vietnamese social media text exhibits significant orthographic variation that challenges tokenization and understanding:

1. **Missing Tone Marks:** Informal writing often omits diacritics: “*buon*” instead of “*buồn*,” “*tram cam*” instead of “*trầm cảm*.”
2. **Abbreviated Forms:** Common abbreviations: “*ko*” = “*không*,” “*dc*” = “*được*,” “*j*” = “*gi*,” “*vs*” = “*với*.”
3. **Expressive Lengthening:** Emphasis through character repetition: “*buônnnnnnn*,” “*mệttttttttt*.”

Implications for Future Work:

These error patterns suggest several directions for improvement:

- Develop disambiguation models that consider full sentence context
- Create specialized Vietnamese tokenizers for informal text

- Explore implicit depression detection through sentiment analysis
- Incorporate emoji and other paralinguistic features

5.8 BERTopic Analysis

We trained BERTopic on the full 316,401-document corpus. The model found 456 topics with 48.30% outlier rate.

Table 5.7: Depression-related topics found by BERTopic

Topic ID	Top Words	Docs	Interpretation
33	trầm cảm, thuốc, bệnh, bs	749	Clinical depression and treatment
7	bs, giấc, ngủ, thuốc, bác	1,687	Sleep disorders and medical consultation
19	burnout, work, stress, job, mental	1,052	Workplace burnout
14	khóc, nghe, buồn, roi, mưa	1,214	Sadness expression and emotional release
27	thân, giấc, ai, cô, buồn, sống	844	Loneliness and existential distress

Table 5.8: BERTopic results on the Round 6 v2 dataset

Model	In Acc	In F1-M	In F1-Dep	X Acc	X F1-M	X F1-Dep
BERTopic-only	0.4449	0.4208	0.3025	0.5078	0.5030	0.5520
PhoBERT + BERTopic	0.8016	0.7153	0.5584	0.5084	0.3674	0.0688

Key Findings: BERTopic-only model shows higher cross-domain F1-macro (0.5030) than in-domain (0.4208), suggesting topic-based features transfer well across domains.

5.9 Domain-Adaptive Pretraining

We ran a controlled experiment on domain-adaptive pretraining (DAPT) using PhoBERT. DAPT gives a small but consistent in-domain gain (+0.0122), but a small cross-domain loss (-0.0107); neither difference is statistically significant.

The cross-domain gap stays large (-0.50 F1 for both models), and was mostly caused by the data itself, not the model. Adapting the encoder alone cannot fix the data-related bottleneck.

5.10 Data Augmentation

An earlier data augmentation experiment showed:

- PhoBERT gained +9.38 F1 points in-domain
- PhoBERT + BERTopic showed the biggest cross-domain improvement (+12.85 F1 points)
- BERTopic-only barely changed (+0.03 F1)

Data augmentation is highly effective for transformer-based models under minority-class imbalance. Future work could explore neural paraphrasing (back-translation, LLM-based paraphrasing).

5.11 Summary of Key Findings

- **Best Model:** PhoBERT with majority voting achieves F1-macro of 0.7187 (in-domain) and 0.3608 (cross-domain)
- **Generalization Gap:** 0.3579 gap due to different label definitions, text lengths, writing styles, and class imbalance
- **Ablation Findings:** Class weighting improves F1 by +6.6 to +12.5 points; active learning shows diminishing returns after Round 3

- **TF-IDF Surprising:** Traditional methods perform competitively (0.7025-0.7138 in-domain) and even better cross-domain (0.3798 for LinearSVC)
- **BERTopic Limitation:** Topic features alone are insufficient (0.4208 in-domain) but transfer better across domains (0.5030 cross-domain)
- **DAPT Counter-Experiment:** Domain-adaptive pretraining shows small, non-significant gains
- **Data Augmentation:** Shows promise for both in-domain (+9.38 F1 points) and cross-domain (+12.85 F1 points for PhoBERT + BERTopic)

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This project makes six main contributions to Vietnamese mental health NLP research:

Contribution 1: VietDepression-125K Dataset A dataset of 125,329 Vietnamese YouTube comments with depression-related keywords, cleaned and preprocessed according to our multi-step pipeline.

Contribution 2: VietDepression-6K Gold Standard A 1,607-sample gold standard dataset with independently verified labels through blind human annotation, achieving Cohen’s Kappa of 0.63.

Contribution 3: Weak Labeling Pipeline A 335-keyword three-level scoring system achieving 98% recall and 61% precision.

Contribution 4: Active Learning Protocol A six-round active learning protocol with selective annotation on uncertain samples.

Contribution 5: Comprehensive Benchmarks Benchmark results for five model families on both in-domain and cross-domain test sets.

Contribution 6: Generalization Gap Analysis In-depth analysis of the 0.3579 F1 generalization gap.

6.2 Key Findings

PhoBERT is the Best Single Model PhoBERT with majority voting across three seeds achieves the best in-domain F1-macro of 0.7187.

Traditional Methods are Competitive TF-IDF + LinearSVC achieves competitive in-domain performance (0.7025 F1-macro) and even better cross-domain performance (0.3798 F1-macro).

The Generalization Gap is the Main Bottleneck The 0.3579 F1 gap is the most significant challenge:

- Different label definitions (distress vs. emotion)
- Different text lengths (long comments vs. short sentences)
- Different writing styles (informal vs. formal Vietnamese)
- Class imbalance effects on balanced test sets

Class Weighting is Essential Class weighting improves F1-macro by +6.6 to +12.5 percentage points across models.

Topic Features Transfer Well but are Insufficient BERTopic-only achieves 0.5030 F1 cross-domain but only 0.4208 in-domain.

6.3 Practical Implications

For Researchers:

- **Dataset papers are valuable:** Our VietDepression dataset provides a foundation for future Vietnamese mental health NLP research.
- **Cross-domain evaluation is essential:** In-domain results can be misleading; always evaluate on held-out domains.
- **Active learning works:** Selective annotation on uncertain samples is more efficient than random sampling.

For Practitioners:

- **Deployment considerations:** A real-world deployment would need domain-specific fine-tuning.
- **Trade-offs exist:** PhoBERT is best for accuracy, but TF-IDF is faster and more generalizable.

- **Human oversight remains essential:** Automated screening should augment, not replace, professional assessment.

6.4 Limitations

This research has several limitations:

- **Platform limitation:** YouTube comments may not represent the broader Vietnamese population's mental health expressions.
- **Language limitation:** Informal Vietnamese spelling variations affect model performance.
- **Annotation limitation:** Cohen's Kappa of 0.63 indicates substantial but not perfect agreement.
- **Generalization limitation:** The large cross-domain gap suggests limited transferability.
- **Clinical limitation:** This research focuses on text classification, not clinical diagnosis.

6.5 Future Work

Dataset Expansion:

- Collect data from multiple Vietnamese platforms (Facebook, TikTok, Zalo).
- Include more demographic diversity in annotated samples.
- Develop a longitudinal dataset to study depression trajectories.

Model Improvements:

- Explore domain-adaptive pretraining (DAPT) with more epochs.
- Investigate large language models (LLMs) like GPT-4 for zero-shot detection.

- Develop multilingual models for cross-lingual transfer.
- Implement neural data augmentation using paraphrasing and back-translation.

Clinical Translation:

- Collaborate with mental health professionals to validate the screening tool.
- Develop severity estimation beyond binary classification.
- Implement early warning systems for at-risk individuals.

6.6 Conclusion

This research presents the first comprehensive study on depression detection from Vietnamese social media text using deep learning models. We constructed the VietDepression dataset with 125,329 YouTube comments and 1,607 gold-standard annotated samples, providing a valuable resource for Vietnamese mental health NLP research.

Our experiments with five model families reveal that PhoBERT achieves the best in-domain performance (F1-macro: 0.7187), while TF-IDF + LinearSVC shows the best cross-domain generalization (F1-macro: 0.3798). The 0.3579 F1 generalization gap highlights the main challenge: the difficulty of transferring models across different platforms and label definitions.

Despite the challenges, this research demonstrates the feasibility of automated depression detection from Vietnamese social media text, paving the way for early intervention tools and mental health support systems for the Vietnamese population.

*The ultimate goal is not to replace mental health professionals,
but to augment their ability to reach and help those in need.*

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APPENDIX

Appendix A. YouTube Search Keywords (Thematic Grouping)

Our 264 search keywords are grouped into five themes, covering a comprehensive range of depression-related topics in Vietnamese:

Group 1: Depression and Suicidality (30 terms) trầm cảm (depression), tôi bị trầm cảm (I have depression), dấu hiệu trầm cảm (signs of depression), vượt qua trầm cảm (overcoming depression), sống với trầm cảm (living with depression), muốn chết (want to die), nghĩ đến cái chết (thinking about death), ý định tự tử (suicidal ideation), tự tử (suicide), đừng tự tử (don't commit suicide), phòng chống tự tử (suicide prevention), cứu tôi với (save me), cần được giúp đỡ (need help), trầm cảm sau sinh (postpartum depression), trầm cảm tuổi dậy thì (adolescent depression), trầm cảm ở người trẻ (depression in young people), sinh viên trầm cảm (depressed university student), học sinh trầm cảm (depressed student), người trẻ trầm cảm (depressed youth).

Group 2: Loneliness and Sadness (42 terms) cô đơn (lonely), cảm thấy cô đơn (feel lonely), nỗi cô đơn (the loneliness), tâm sự đêm khuya (late-night confessions), tâm sự buồn (sad confessions), buồn không lý do (sad for no reason), cảm xúc tiêu cực (negative emotions), năng lượng tiêu cực (negative energy), khóc một mình (crying alone), nước mắt đêm khuya (late-night tears), một mình trong phòng (alone in the room), nụ cười che giấu nỗi buồn (smile hiding sadness).

Group 3: Stress and Burnout (45 terms) mệt mỏi cuộc sống (tired of life), chán nản cuộc sống (bored with life), không còn động lực (no motivation left), stress công việc (work stress), áp lực công việc (work pressure), burnout, kiệt sức công việc (burnout at work), áp lực học tập (academic pressure), stress học đường (school stress), áp lực gia đình (family pressure), áp lực tài chính (financial pressure), lo âu (anxiety), rối loạn lo âu (anxiety disorder).

Group 4: Sleep Disturbance and Overthinking (18 terms) mất ngủ (insomnia), mất ngủ vì stress (insomnia due to stress), đêm không ngủ được (can't sleep at night), khó ngủ (difficulty

sleeping), rối loạn giấc ngủ (sleep disorder), overthinking ban đêm (nighttime overthinking), suy nghĩ quá nhiều (thinking too much), nghĩ nhiều mệt mỏi (tired from overthinking).

Group 5: Psychological Help-Seeking and Healing (30 terms) tư vấn tâm lý (psychological counseling), trị liệu tâm lý (psychotherapy), chữa lành tâm lý (psychological healing), chữa lành đứa trẻ bên trong (healing the inner child), yêu bản thân (self-love), vượt qua nỗi đau (overcoming pain), nhạc chữa lành (healing music).

Appendix B. Detailed Hyperparameter Configurations

Table 6.1: Training and hyperparameter settings for PhoBERT, BiLSTM, and TF-IDF + SVM.

Parameter	PhoBERT	BiLSTM	TF-IDF + SVM
Batch size (train / eval)	8 / 16	32	—
Epochs	3 (patience=2)	8	—
Learning rate	2e-5	1e-3	—
Optimizer	AdamW	Adam	—
Weight decay	0.01	—	—
Warmup ratio	0.06	—	—
Gradient clip norm	1.0	—	—
Max sequence length	128	100	—
Class weighting	Computed	—	balanced
Embedding dim	768	128	—
Hidden dim	768	256	—
LSTM layers	—	2	—
Dropout	—	0.5	—
Vocabulary size	—	15,000	—
N-gram (word)	—	—	1-2
N-gram (char)	—	—	3-5
Max features	—	—	80,000
Solver	—	—	LinearSVC
Max iterations	—	—	2,000
Random seed	42	42	42

Appendix C. Illustrative Error Examples

Key Observations:

Table 6.2: Typical error examples. FP = false positive, FN = false negative, TP = true positive.

Comment (VN)	English gloss	True	Pred	Error Type
"buồn ngủ quá trời ơi"	so sleepy, oh my god	normal	depression	FP: "buồn" = sad/sleepy
"mình mệt mỏi quá mn ạ"	I'm so tired, everyone	normal	depression	FP: physical tiredness
"chẳng biết bao giờ mới ổn"	don't know when I'll be okay	depression	normal	FN: hidden distress
"lại một ngày nữa trôi qua"	another day passes by	depression	normal	FN: low-interest tone
"có ai như mình ko, ngày nào cũng thấy trống rỗng"	feeling empty every day	depression	depression	TP: "trống rỗng" (empty)

- **False Positives** often occur when Vietnamese words have multiple meanings (e.g., "buồn" means both "sad" and "sleepy")
- **False Negatives** typically involve implicit expressions of distress without explicit depression keywords
- **True Positives** demonstrate that explicit keyword matching works well when the signal is clear